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Who Merges with Whom? EPS Accretion and the Market for Corporate Control♣

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Abstract: We examine the role of expected EPS effects (possible accretion/dilution) in shaping merger activity. We find robust evidence that the number of EPS-accretive bidders – buyers that could increase earnings per share when buying the target – is a strong predictor of both overall merger activity and the likelihood of a firm becoming a target. Similarly, the number of EPS-accretive targets predicts which firms are likely to become acquirers, especially when the acquirer's executive pay is more sensitive to earnings. The effect of accretion opportunities on takeover likelihood was substantially larger before the 2001 elimination of pooling-of-interests accounting, when acquirers could more easily preserve reported earnings. Our results suggest that EPS incentives remain central in explaining merger activity and who merges with whom. The effect is specific to earnings: comparable counts based on book value have little incremental predictive power. It is also concentrated among large acquirers.

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1. Introduction

Mergers and acquisitions are perhaps the most important investments made by firms on behalf of their investors. While these transactions are presumably motivated by synergy gains and the associated increase in future cash flows, almost all merger announcements made by public acquirers include some reference to earnings per share (EPS) accretion or dilution associated with the transaction.[1] Investor concerns over EPS dilution are particularly acute to the extent that a transaction results in the issuance of new shares of stock as compensation to target shareholders. Indeed, Dasgupta, Harford and Ma (2024) find that acquirer's EPS focus affects how deals are structured and the premium that is paid.

However, existing evidence on EPS 'bootstrapping' yields mixed conclusions regarding its value to shareholders. That is, while Dasgupta, Harford and Ma (2024) explore the impact on the deals that occur, we are not aware of research that explores whether EPS considerations are important in understanding whether a deal occurs in the first place: i.e., overall merger activity and the matching of bidders and targets. While Andrade (1999) finds that EPS accretion has a positive and statistically significant effect on acquirer abnormal returns, he concludes that the economic effect is too small to justify a manager's concern.[2] Despite seemingly cosmetic EPS effects, the attention placed on merger accretion by market participants suggests that managers may indeed factor in the expected impact on earnings when deciding whether to engage in a merger, and if so, with whom. This incentive may be important given a couple of related factors. The price-earnings

[1] For example, in announcing its $20 billion acquisition of Nuance Communications in 2021, Microsoft executives emphasized that the transaction would be minimally dilutive to the firm's 2022 non-GAAP earnings, and accretive to earnings in the year after. https://news.microsoft.com/2021/04/12/microsoft-accelerates-industry-cloud-strategy-for-healthcare-with-the-acquisition-of-nuance/

[2] Brealey et al. (2006, chapter 32) classifies EPS accretion as one of the "dubious" reasons for mergers. Bruner (2004, chapter 16) states "Earnings per share (EPS) is an ongoing concern to executive and directors who believe that this single measure is the main focus of attention by investors and the financial community." In a related footnote he reports "Scholarly research is at best of mixed agreement with this view; some research is hostile to it."

(P/E) ratio is known to be an important valuation ratio for market participants, and managerial compensation can be economically sensitive to earnings. Managers also take costly real actions to meet earnings targets: firms that would narrowly miss analysts' EPS forecasts repurchase shares to close the gap, at the expense of investment and employment (Almeida, Fos, and Kronlund, 2016), and salient but economically irrelevant reference points such as the 52-week high measurably shape takeover pricing (Baker, Pan, and Wurgler, 2012). Indeed, if such incentives are powerful enough, EPS may be an important consideration in overall merger activity and the matching of bidders and targets.

This study also considers how acquisition incentives are affected by the relative valuation of bidders and targets. It is now widely accepted that some acquisitions are driven by stock market misvaluation rather than synergies or agency conflicts (e.g., Shleifer and Vishny, 2003; Rhodes-Kropf and Viswanathan, 2004; Jensen, 2005). Bidders are generally more highly valued than their targets (Dong et al., 2006), and both parties tend to be in temporarily high-valued industries (Rhodes-Kropf et al., 2005). As relative valuation can also be an important factor in determining bidder and target matching in the takeover market, as well as transaction characteristics, our tests also account for perceived "misvaluation". While our EPS-based view is related to these valuation-based explanations, it differs in that we focus specifically on EPS-accretion/dilution incentives.

We start by examining whether EPS considerations are important in determining overall merger activity and the matching of bidders and targets. To address this issue, we consider the universe of prospective mergers between any two firms in the same two-digit SIC industry. We then classify bidders and targets as *accretive* if, by using stock as the entire method of payment, a potential bidder can increase its EPS after paying a reasonable premium. We then calculate the total number of accretive bidders and targets for each firm each year (in an industry). The resulting

measures broadly resemble the dispersion of valuation multiples within an industry as illustrated in Harford (2005)—indeed, aggregated to the industry level, the number of accretive pairs is itself a threshold-based measure of P/E dispersion—but they are grounded in a specific model of managerial behavior (EPS focus) and, empirically, outperform conventional dispersion measures such as the standard deviation and interdecile range of valuation multiples.

Our evidence suggests that overall merger activity is closely tied to EPS accretion opportunities. In industry years where there are more accretive bidders, then there is significantly higher merger activity. Merger activity is also higher in years where there were more accretive bidders in the *prior* year. This is true after controlling for other factors known to predict merger activity, as well as prior year merger activity and overall merger activity within the industry.

After examining overall activity at the industry level, we turn next to the matching of bidders and targets. We find that a firm is more likely to be a target if it has more accretive bidders in its industry. This is robust to other factors known to impact whether a firm is a target, including size and valuation metrics. We next find that a firm is more likely to be a buyer if it has more accretive targets in its industry. This result is also robust to standard controls when we include year and industry fixed effects. Because whether one firm is an accretive bidder for another depends only on the two firms' price-to-forecast-earnings ratios, both counts are closely related to a firm's P/E rank within its industry. Our counts therefore cannot be fully separated from relative valuation, and we place most weight on patterns that a pure relative-valuation explanation does not predict: the concave relation implied by our model, the concentration of the effect among large acquirers and acquirers with earnings-sensitive pay, and its variation around the elimination of pooling-of-interests accounting.

Our tests assume bidders issue new shares. Survey evidence in Graham and Harvey (2001) suggests that managers of large, regulated and dividend-paying firms are concerned with EPS dilution when issuing new equity. Consistent with this evidence, the sensitivity of acquisition likelihood to accretion opportunities is significantly greater for larger firms; we find no reliable differences for regulated or dividend-paying firms.

We exploit a natural experiment — the 2001 accounting rule change — to identify variation in the supply of EPS-motivated buyers. Before mid-2001, an acquirer could choose between the pooling of interest and purchase methods in merger accounting.[3] The pooling method enables the buyer to report combined earnings after the deal, essentially adding financial statements together at book value, while the purchase method marks assets and liabilities to fair market value and records any additional premium paid as goodwill. Prior to 2001, goodwill was amortized over time, reducing earnings in each period. The increased basis from marking other assets to market likely also served as a drag on future earnings. Thus, a buyer concerned about EPS will generally prefer pooling. SFAS 141 eliminated pooling-of-interests accounting for business combinations initiated after June 30, 2001, and SFAS 142 replaced goodwill amortization with an annual impairment test, which may trigger a write-off reducing future EPS. Thus, a hypothetical EPS accretion deal according to our experiment should be less attractive in this later period. Indeed, excluding the 2001–2002 transition years, the effect of accretive bidders on takeover likelihood is substantially larger in the decade before the change than in the decade after.

We also examine two channels suggested by prior work. If managers pursue accretive deals because their pay is tied to EPS (e.g., Grinstein and Hribar, 2004; Dasgupta, Harford and Ma,

[3] Strictly speaking, pooling was not a free choice, but acquirers could structure deals in a way that allowed for pooling-of-interest accounting, and firms sometimes went to great lengths to do so (see, for example, Lys and Vincent (1995)).

2024), accretion opportunities should matter more for acquirers whose executives have earnings-sensitive pay. Using the earnings-based pay-performance sensitivities of Bennett et al. (2017), we find evidence that they do: the relation between the number of accretive targets and the likelihood of becoming an acquirer is significantly stronger when the firm's own pay-performance sensitivity is high. For a firm at the 75th percentile of that sensitivity, moving from the 25th to the 75th percentile of accretive targets raises the annual probability of an acquisition by about 0.9 percentage points; for a firm at the 25th percentile, the change is essentially zero. If financing frictions bind, accretion opportunities should also matter more when potential bidders are cash-rich (Harford, 1999); we find no reliable evidence that they do.

To assess whether our results reflect earnings specifically rather than relative valuation more broadly, we construct analogous counts of viable bidders in which book value per share or residual-income intrinsic value replaces forecast EPS in the accretion rule. When the three counts compete, the EPS-based count retains a large and significant coefficient, the book-value count has little incremental predictive power, and the intrinsic-value count—which is itself built from analyst earnings forecasts—has a smaller but significant effect.

This paper makes several important contributions to the literature. First, we use a unique matching algorithm to identify likely bidders and targets that relies on prospective EPS accretion. The existing literature on target prediction largely focuses on target attributes with limited success.[4] More notably, matching in the market for corporate control likely depends on valuation metrics relevant to managers and investors such as earnings, as well as relative (mis)valuation as

[4] See for example Hasbrouck (1985), Palepu (1986), Mikkelson and Partch (1989), Harford (1999), and Cremers et al. (2009). For reference, specifications in Cremers et al. (2009) use an extensive panel of 83,000-plus firm-year observations and incorporate a battery of firm specific characteristics including growth opportunities, capital, and ownership structure. The explanatory power of their models is approximately 3%.

documented in Rhodes-Kropf et al. (2005). In addition, our study contributes to a growing literature on matching between financial market participants. Although our paper differs from the papers that study matching in corporate finance generally – for example, Sørensen (2007), Schwert (2018), and Akkus et al. (2021) – our methodology is essentially to match firms based on outcomes that we hypothesize are important to bidders, namely EPS accretion/dilution. As such, our paper can be seen as an extension of the structural model of Akkus et al. (2016) that assumes observed matches in the takeover market maximize synergies.

Second, we exploit the 2001 elimination of pooling-of-interests accounting as a natural experiment to identify variation in the supply of EPS-motivated buyers. The substantially stronger effect of our accretion measure before pooling was eliminated—when its accounting benefits made EPS accretion especially salient—is consistent with an accounting earnings channel, although the two periods also differ in other respects. Third, our analysis builds directly on Dong et al. (2006), extending their valuation-matching framework to ex ante bidder and target identification. They study completed acquisitions and find strong evidence that buyers are more highly valued than their targets based on either book or fundamental value. Our paper focuses on the broader issue of predicting *a priori* which firms ultimately become acquirers or targets. Finally, we show that the link between accretion opportunities and acquisitions is concentrated among large acquirers, consistent with survey evidence that concern over EPS dilution is greatest at large firms (Graham and Harvey, 2001), and that it is stronger for acquirers whose executive pay is more sensitive to earnings, linking EPS-motivated acquisitions to managerial incentives (Grinstein and Hribar, 2004; Dasgupta, Harford and Ma, 2024).

The remainder of the paper is organized as follows. In Section 2 we motivate our empirical approach in the context of misvaluation. We then describe our approach for determining the

number of viable buyers and viable targets for each firm and the corresponding predictions. Section 3 describes our data sample and methodology. We present our empirical results in Section 4 and conclude in Section 5.

2. Relative Valuation and Viable Bidders and Targets

Shleifer and Vishny (2003) develop the first model in which acquirers use overvalued stock as currency to buy relatively less overvalued targets. The target benefits in the short term by getting a premium in the deal. The acquirer benefits in the long term by getting a larger share of the combined firm than they can get if both firms are valued at their long-run values. Jensen (2005) argues that overvalued equity may lead to bad acquisitions, which reduces the core values of the firms and results in poor long-term operating performance. This is similar to the Shleifer and Vishny (2003) model, except in this case it is managers who mistakenly believe that the market will use the acquirer's multiple for the combined entity. In Appendix B we show how either of these views of bidder behavior can be transformed into a model predicting the likelihood of a bidder and a target consummating a merger.

Rhodes-Kropf and Viswanathan (2004) (hereafter RKV) also offer a model based on market misvaluation, but targets are not myopic as in Shleifer and Vishny (2003). Instead, the value of both targets and acquirers can deviate from their long-run fundamental value. RKV separate this stock misvaluation into a market-wide (or sector) component and a firm component. The target CEOs cannot determine how much of this misvaluation is due to firm-specific reasons rather than market or sector misvaluation. The target CEO correctly adjusts downward the stock offer received from the acquirer but still accepts the offer because he overestimates the merger synergies owing to the common misvaluation of the market or sector.

Rhodes-Kropf et al. (2005) test the RKV and Shleifer and Vishny (2003) models using an empirical valuation model that includes book values, net income and leverage to decompose the market value mispricing into three components. The three components of their decomposition are a firm-specific error, a sector mispricing error and a long-run mispricing error. They find that both targets and acquirers have a higher market-to-book (M/B) compared to non-merger firms, and high M/B targets are bought by even higher M/B acquirers. The firm-specific error is higher for acquirers than targets in the overall merger sample and for the stock-financed sample. However, they also find that low long-run, value-to-book firms acquire high long-run, value-to-book targets. The empirical work of Rhodes-Kropf et al. (2005) explains a significant amount of overall merger activity, medium of exchange, and bidder identification; we include the components of their decomposition as controls in Table 6 and, at the industry level, in Table 4.[5]

Our approach yields two testable predictions. First, holding other characteristics fixed, a firm is more likely to become a target the more firms in its industry could acquire it in an EPS-accretive, all-stock transaction at a conventional premium. Second, a firm is more likely to become

[5] There is an extensive literature that focuses on misvaluation and market timing in the context of mergers. Ang and Cheng (2006) use a similar P/B and P/V approach and have findings analogous to those in Dong et al. (2006). Further, they show that the shareholders of acquirers in stock mergers are at least as well off as the shareholders of similarly overvalued non-acquiring firms. Moeller et al. (2005) find that the negative bidder returns from 1998 to 2001 are driven by a few deals in which the bidders with extremely high values suffer huge losses after merger. They find that such firms have high q's and high market-to-book ratios. Thus, this provides support for Fuller and Jensen's (2002) claim that managers of high valuation firms make poor acquisitions. However, the negative returns can also be due to the market's adjustment of the true stand-alone values of such firms. Song (2007) uses the trading by insiders as an indication of overvaluation. She finds a strong relation between the insider selling prior to the merger announcement and long-run post-acquisition performance in the "hot market" period of 1997-2000. Bouwman et al. (2009) find that acquisitions in high valuation periods generate significantly lower long-term abnormal stock returns and operating performance for the buyers. However, they show that market timing is not likely an explanation for the underperformance of acquirers in high valuation market. Fu et al. (2013) investigate whether acquirer shareholders benefit from acquisitions driven by equity overvaluation. They sort acquisitions by relative overvaluation before the announcement. Their findings support Jensen's (2005) hypothesis that equity overvaluation generates substantial agency costs for shareholders. Bi and Gregory (2011) use UK data and find more support for the market misvaluation hypothesis than the Q theory; however, they cannot comprehensively reject the Q theory explanation.

an acquirer the more firms in its industry it could acquire on an accretive basis. Because only one willing bidder is required for a takeover, the simple model in Appendix B further implies that the probability of a takeover is a concave function of the number of accretive bidders, rather than a linear function of a firm's valuation rank.

3. Sample and Empirical Methodology

In this section, we summarize our data and sample selection, then define our variables of interest and our controls. We close by summarizing descriptive statistics, and present univariate results related to the number of viable buyers and the likelihood of a firm being a takeover target.

3.1 Data

Our sampling incorporates all firms listed in the Compustat universe. We supplement firm-level data with merger and acquisition transaction data drawn from SDC Platinum dataset. We download all deals with domestic targets from the SDC database from 1980 to 2022. The search includes all transactions in four categories: "Merger", "Acquisition of Majority Interest", "Acquisition" or "Acquisition of Assets". We only consider deals in which the acquiror is acquiring an interest of 50% or more in a target to ensure that buyers can integrate (or consolidate) the targets' financials into their own. Appendix C details our SDC sample construction, Appendix D provides variable definitions, and Appendix A reports additional robustness tables.

Of the 3,959 target firm-years in our final sample, 3,859 are mergers, 64 are acquisitions of majority interest and 36 are acquisitions of assets.

To calculate forecasted EPS-based numbers of viable buyers and targets, we use I/B/E/S earnings annual forecast data. We use the one year (fiscal) average EPS forecast in I/B/E/S as the forecasted EPS for the firms in the year.

3.2 Empirical Specification and Key Variable

We estimate the following Probit regression model for merger likelihood:

$$y_{i,t} = f(\alpha + \beta_1 V_{i,t-1} + \beta_2 X_{i,t-1} + \beta_3 Z_t + \beta_4 W_{t-1} + \mu_t + \nu_j) \quad (1)$$

where i denotes firms, t refers to time in years, j denotes industry, μ_t refers to time fixed effects and ν_j refers to industry fixed effects. The dependent variable y is an indicator variable for a merger. In the regression of target likelihood, y is equal to 1 if firm i is a target in year t and 0 otherwise. The corresponding V is the number of viable buyers for firm i at year t-1. X's are the control variables, including firm size and its square, leverage and the price-to-forecast-earnings ratio. Z includes the number of firms in the industry and the dispersion (standard deviation or interdecile range) of the price-to-forecast-earnings ratio within the industry. W is the number of acquisitions that took place in firm i's industry j.

3.3 Test Variables

To compute the number of viable buyers for each firm, we assume that any potential buyer will pay for the target with its own equity. As the issuance of additional equity will dilute any forward-looking EPS accretion attributable to a hypothetical combination, this assumption places a conservative lower bound on prospective deal accretion. For a given firm, in each year, we compute the number of firms that are able to make an equity-financed deal that is earnings per share accretive while paying a 20% acquisition premium to the target. For example, we identify firm B as a viable buyer for firm A if firm B uses its stock, pays a 20% premium for firm A's equity (in market value), *and* the resulting prospective earnings per share of firm B increases after absorbing firm A. We denote the number of viable buyers based on EPS as *Accretive Bidders*, and the number of viable targets there are for each bidder as *Accretive Targets*. Both the focal firm and the potential counterparty must have positive forecast EPS, a positive stock price and positive

shares outstanding, and each count includes distinct firms in the same two-digit SIC industry-year, excluding the firm itself. Firm-years with negative or zero forecast EPS are therefore excluded from the counts; Appendix Tables A5 and A6 show that our results are similar when negative forecast EPS is retained.

3.4 Control variables

We utilize a wide range of controls from existing literature. Past studies (Hasbrouck, 1985; Palepu, 1986; Mikkelson and Partch, 1989) show that the likelihood of being a target is negatively related to the size of the firm. We use various size measures as robustness checks, but settle on log of assets and its square for the results reported here. Prior target-prediction models (e.g., Palepu, 1986) include the price-to-earnings ratio, so we also control for it. Since Cremers et al. (2009) find that leverage has a positive and significant effect on being a takeover target, we also include leverage, measured as the book value of debt divided by total assets, as a control. Finally, as our paper is most closely related to Rhodes-Kropf et al. (2005), we adopt their measures as key control variables. Following their method, we decompose the market-to-book ratio into three components, including firm-specific errors, industry time series errors, and long run value-to-book errors. We then verify that our decomposition method obtains similar results as those reported in Table 9 of their paper.[6] Lastly, we use the same set of firm financial variables in the regressions related to the likelihood of being an acquirer as we expect such variables should have opposite effects.[7] Our

[6] Past research has also used a host of alternative valuation metrics. For example, Hasbrouck (1985) argues that low Market/book values can proxy for management incompetence and low-cost resources for acquirers. Cremers et al. (2009) find that a firm's q is negatively related to the probability of being a target. They also find that ROA (return on assets) is negatively related to the likelihood of being a target, which we measure as the ratio of net income before extraordinary items and discontinued operations to total assets. Hasbrouck (1985) also notes that it is easier for a bidder to secure a toehold in a more liquid target. In unreported tests, we add these variables as controls, and their inclusion does not change any of our primary conclusions. Since the Rhodes-Kropf et al. valuation decomposition overlaps with most of these metrics, in the paper we do not include them in our main specifications.

[7] We winsorize the ratios at the 1% level.

baseline target and acquirer regressions do not include valuation measures beyond the price-to-forecast-earnings ratio; in Table 6 we add the three components of the Rhodes-Kropf et al. (2005) decomposition (their Model 2), and our results are robust to their inclusion.

Next, we sort the data by two-digit SIC code and calendar year. Following the merger wave literature, we compute *Ind Num Takeover* as the total number of takeovers in the industry in the year. We also include year and industry fixed effects and cluster standard errors at the industry level. It is widely understood that mergers are more commonly observed in booming markets so we control for market performance.

To ensure that our results are not driven by an industry effect, we control for industry dispersion (standard deviation and interdecile range) of the price-to-forecast-earnings ratio. Because the number of viable buyers is mechanically related to the number of firms in the industry, we include the number of firms in the industry as an additional control.

3.5 Summary Statistics

Table 3 offers a first look at whether our theory is plausible. Using one observation per year from 1980 to 2022, we regress the number of takeovers on the median number of accretive bidders in the current and prior year, controlling for the level of the S&P 500 and market-to-book dispersion. Takeover activity is strongly related to the median number of accretive bidders both contemporaneously and with a one-year lag, and these simple regressions explain 58% to 62% of the annual variation in takeover counts.

We present the summary statistics of our key variables in Table 1, Panel A. Here, focusing on lagged values of each, Acquirer Sample includes all firms that were an acquirer in a particular

year, Target Sample includes all firms that were a target in a particular year, and Non-Merger Sample includes the set of firms not involved in merger activity. As is typical, non-merger firms account for more than 90% of the observations. Our sample contains 91,547 firm-years from 1980 to 2022: 3,992 acquirers, 3,959 targets and 83,596 non-mergers. Most of the reduction from the linked SDC sample reflects the requirement of a prior-year analyst EPS forecast and positive forecast EPS; Appendix C details each step and reports sample coverage by decade. Note that the means and medians of viable buyers (and viable targets) are quite different, and the distribution of viable bidders (and viable targets) is highly right-skewed. Hence, following Aggarwal and Samwick (1999), we use the empirical cumulative distribution function (CDF) to normalize the independent variables and run our regressions with these transformed variables.

Table 1, Panel B provides univariate comparisons for our key variables across the bidder, target and non-merger samples. Bidders compared to non-mergers (all variables are lagged) have higher industry number of takeovers, are larger (assets), have less leverage, have higher industry number of firms, and no significant difference in lagged Price/Forecasted Earnings. Bidders also have more accretive targets and more accretive bidders.

The target sample is also very different than the non-merger sample for our key variables. Targets have higher industry number of takeovers, are smaller (assets), have lower leverage, have higher industry number of firms, and have a lower price/forecasted earnings. They also have more accretive bidders. For example, firms in the target sample have a mean of 65 viable buyers (*Lagged Accretive Bidders*), whereas those in the *Non-Merger* sample have just 45.

Comparing bidders with targets largely restates their differences from non-mergers described above. The number of accretive bidders is similar across the two groups (means of 65 for targets and 60 for acquirers), although the difference is statistically significant. Although

univariate comparisons are only suggestive, they indicate that the number of accretive bidders is a strong determinant of merger activity. However, whether the firm is a bidder or a target is then determined by other factors. Bidders are in industries with more takeovers, are larger, have lower leverage, have more firms in the industry, have higher price/forecasted earnings, and have more accretive targets. Many of these characteristics are known predictors of bidders (e.g., Rhodes-Kropf et al., 2005); what is new in our analysis is the role of EPS accretion.

Panel C shows a correlation matrix for our key variables. An important observation is the high correlations between the number of takeovers by industry and each of our measures of the number of viable bidders and targets. One reason is that while our model is focused on individual firms as targets, we use industry to restrict our search for potential buyers. Our multivariate results are similar using Fama–French 48 industries (Appendix Tables A5 and A6), and our measures predict mergers at the industry level as well as the firm level (Table 4). Panel D correlates the industry-year median of each lagged count with the industry-year merger rate, weighting each firm-year equally. The Spearman correlations are about 0.37 for both Accretive Bidders and Accretive Targets; Pearson correlations are lower, about 0.33 to 0.35, reflecting the skewness of merger rates. Book- and RIM-based counts have similar univariate correlations, so the multivariate tests below are needed to distinguish among them. Unlike the merger-wave approach, which cannot make a prediction until actual mergers have been observed, our measures are available before any deals occur.

4 Main Empirical Results

In this section, we present multivariate evidence on our primary hypotheses. Tables 2 through 4 focus on overall merger activity, Tables 5 through 7 on targets and acquirers, and Tables 8 through 11 on cross-sectional and time-series variation that bears on the mechanism behind our

results. Appendix A reports robustness tests. Unless otherwise noted, all regressors are lagged one calendar year, continuous regressors are transformed using their annual empirical CDFs, coefficients are probit index coefficients, and standard errors are clustered by two-digit SIC industry.

4.1 Predicting Merger Activity at Firm and Industry Levels

In Table 2, we provide the results of a probit model on our ability to predict overall merger activity. The dependent variable is a merger dummy variable equal to 1 if a firm is a takeover target or a buyer. The independent variables (except dummy variables) are transformed by empirical cumulative distribution function (CDF) per our discussion above. We cluster the standard errors at the industry level. This model only includes our key independent variables with and without fixed effects. These parsimonious models provide a first look before we add other factors known to explain merger activity. The estimated coefficients on Accretive Bidders and Accretive Targets are positive in every specification and significant in all but one (Accretive Targets with industry fixed effects only). As discussed earlier, given skewness in our key variables, these models use the empirical cumulative distribution function (CDF) to normalize the independent variables. In unreported results we also estimated OLS models with similar results.

In Table 4, we investigate whether our measures also predict merger activity at the industry level. The dependent variable is the number of merger announcements in each two-digit SIC industry and year, and the key predictive variable is the industry mean of the lagged number of accretive bidders (Industry mean accretive bidders). Additional controls include the number of firms in the industry; the industry mean market-to-book ratio or, in columns 7 to 10, the industry means of the Rhodes-Kropf et al. (2005) time-series and long-run valuation components; and the number of takeovers in the prior year, both economy-wide and within the industry.

Industry mean accretive bidders is positive and significant at the 1% level in every specification. Controlling for the number of firms in the industry or for industry valuation has little effect on the coefficient. Adding the industry's own takeover count in the prior year roughly halves the coefficient (from about 0.13 to 0.07), but it remains highly significant, so accretion opportunities predict merger activity beyond the persistence of industry merger waves (Harford, 2005; Ahern and Harford, 2014).

4.2 Predicting Takeover Targets Based on the Number of Viable Bidders

We now turn to our hypothesis that the likelihood of being a takeover target is increasing in the number of viable bidding firms. Table 5 reports probit regressions in which the dependent variable equals 1 if the firm is a takeover target and 0 otherwise. Columns 1 to 3 include year and industry fixed effects, and columns 4 to 6 omit them. Columns 1 and 4 include only Accretive Bidders; the other columns add controls for the dispersion of the industry's P/FE ratio (its standard deviation or interdecile range), the number of firms in the industry, the firm's P/FE ratio, size and its square, leverage, and the number of takeovers in the industry. Consistent with our primary hypothesis, the coefficient on Accretive Bidders is positive and significant at the 5% level or better in all three columns with fixed effects (0.71, 0.27 and 0.29).[8] Consistent with Rhodes-Kropf et al. (2005), the Price/Forecast Earnings variable is negative and significant, suggesting targets have relatively low relative valuation metrics. Leverage enters positively, consistent with Cremers et al. (2009), and size has an inverted-U relation with takeover likelihood. Without fixed effects, the coefficients are 0.50, 0.26 and 0.23, significant at the 1%, 5% and 10% levels, respectively.

[8] We see that the estimated coefficients for the controls on industry valuation dispersion in columns 2 and 3 are slightly negative and not significant.

To gauge economic significance, note that the unconditional probability of becoming a target in our sample is 4.3%. Because Accretive Bidders is transformed to its empirical CDF, its coefficient measures the effect of moving from the bottom to the top of its distribution. Evaluated at the sample takeover rate, the column 3 coefficient of 0.29 implies that this move raises the annual probability of becoming a target by about 2.6 percentage points, to 6.9%—an increase of roughly 60%; moving from the 25th to the 75th percentile raises it by about 1.3 percentage points. For comparison, Jenter and Lewellen (2015) find that the probability of receiving a bid rises from about 4% to 6% per year when target CEOs reach retirement age. Among the control variables, the firm's price-to-forecast-earnings ratio, size and industry takeover activity are also significant.[9] In unreported results, we run OLS models with the same specifications and find that while our key variable remains positive, it is not significant in all specifications.

A natural concern is that Accretive Bidders simply restates the firm's relative valuation. Whether one firm is an accretive bidder for another depends only on the two firms' price-to-forecast-earnings ratios: with a 20% premium, B is accretive for T exactly when B's P/FE ratio exceeds 1.2 times T's. Within an industry-year, the number of accretive bidders is therefore closely related to the firm's P/FE rank. Because the count is determined by the firm's position in its industry's P/FE distribution, a control for the firm's within-industry P/FE percentile absorbs much of the variation on which our hypothesis rests. In untabulated results, adding such a control leaves the coefficient on Accretive Bidders similar in magnitude but estimated imprecisely. We therefore

[9] Harford (1999) uses the log value of the target's assets as proxy for size and finds that the coefficient is not significant in the prediction of merger targets. Our results are consistent with Harford's. To ensure that size has no effect on our main results, we split the sample according to median firm size. In unreported results, we find that the coefficient for size is negative in the sub-sample of firms above the median size and positive in the sub-sample of firms below the median size, with both significant at the 1% level. This finding points to a nonlinear relationship between size and the likelihood of being a target (Vijh and Yang 2013). Since the effect of size on merger activity is not central to our paper, we add the square of the target firm's size in the regression to control for this nonlinear relationship. With this specification, size has an inverted-U relation with takeover likelihood (Table 5).

rely on features of the evidence that a pure relative-valuation explanation does not predict. First, our model implies that takeover likelihood is a concave function of the number of viable bidders, and the logarithm of the count is significant at the 1% level in every specification, with and without fixed effects (Appendix Table A1); the logarithm of one plus the count is significant at the 5% level or better (Table A2). Second, the relation is concentrated among acquirers for which reported EPS should matter most: large firms (Section 4.5) and firms whose executives' pay is sensitive to earnings (Section 4.7). Third, it varies over time with the accounting treatment of stock-financed acquisitions (Section 4.6).

Overall, the evidence presented in Table 5 provides support for the hypothesis that firms with more viable buyers are more likely to be acquired.

These results are robust to alternative research design choices. Appendix Table A5 re-estimates column 3 of Table 5 with acquisition premiums of 10%, 30% and 40% (columns 1 to 3), with Fama–French 48 industries (column 4), retaining firms with negative forecast EPS (column 5) and clustering standard errors by industry-year (column 6). The coefficient on Accretive Bidders ranges from 0.28 to 0.32 and is significant at the 5% level or better in every case.

4.3 Earnings or Valuation? Accretive, Book and RIM Bidders

Our accretion rule could capture relative valuation in general rather than earnings specifically. To distinguish these possibilities, we construct two analogous counts. Book Bidders replaces forecast EPS in the accretion rule with book equity per share, and RIM Bidders replaces it with a three-year residual-income estimate of intrinsic value per share (Frankel and Lee, 1998; Dechow, Hutton, and Sloan, 1999; Lee, Myers, and Swaminathan, 1999), which is itself built from analyst earnings forecasts (Appendix D gives definitions). Table 6 includes all three counts, together with the three components of the Rhodes-Kropf et al. (2005) valuation decomposition, in

the 62,634 firm-years for which all measures are available. With year and industry fixed effects (columns 1 to 3), Accretive Bidders has the largest coefficient in every specification (0.35 to 0.91) and is significant in all of them, Book Bidders is never significant, and RIM Bidders is positive and significant with a smaller coefficient (0.21 to 0.32). Without fixed effects (columns 4 to 6), the evidence is weaker: Accretive Bidders is significant in the parsimonious model (0.64), but with controls its coefficient falls to 0.10 to 0.16 and is insignificant, while RIM Bidders remains significant and Book Bidders is positive and significant in one specification and negative and significant in another. With fixed effects, the pattern holds with 10%, 30% and 40% premiums, with Fama–French 48 industries, when negative forecast EPS is retained and with industry-year clustering (Appendix Table A6); Book Bidders is significant only with Fama–French 48 industries (at the 5% level) and with industry-year clustering (at the 10% level). In our main specifications, accretion opportunities defined by earnings, rather than by book value, predict which firms become targets.

4.4 Predicting Bidders Based on the Number of Viable Targets

We now turn to predicting acquirers. Table 7 reports probit regressions in which the dependent variable equals 1 if the firm is an acquirer and 0 otherwise, using the same structure as Table 5: columns 1 to 3 include year and industry fixed effects, and columns 4 to 6 omit them. The coefficient on Accretive Targets is 0.24 in the parsimonious model in column 1 and 0.23 to 0.24 with controls in columns 2 and 3, all significant at the 5% level or better, so the likelihood of becoming an acquirer increases with the number of viable targets. Size has a strong positive effect, leverage enters negatively and the number of takeovers in the industry is positive and significant, consistent with existing literature; industry valuation dispersion is insignificant. Without fixed effects, the coefficient is large and significant in the parsimonious model (0.58, column 4) but

small and insignificant once controls are added (0.04 to 0.05). The logarithm of the count—the concave form implied by our model—is significant in every specification, with and without fixed effects (Appendix Tables A3 and A4), and Section 4.5 shows that the linear effect is concentrated among large firms.

4.5 Firm Size, Dividends and Regulated Firms

In this subsection, we explore the implications of managerial attention to EPS on the likelihood of being a bidder. Evidence presented in Graham and Harvey (2001) suggests that concern over EPS dilution is greater among regulated companies, large firms, and those that pay dividends. Hence, if accretion is important for the likelihood of being an acquirer, this effect should be most pronounced for these firms. To test this hypothesis, we create a dummy variable *Regulated* which takes the value of 1 if a firm is in the finance industry (SIC code between 6000 and 6999) or in the utilities industry (SIC code between 4800 and 4914), and 0 otherwise. If a firm pays dividend, then *Dividend* is set to 1 and 0 otherwise for nonmissing dividend data; missing dividend data remain missing. We also create an interaction term between *Size* (*Regulated*, *Dividend*) and *Accretive Targets*. We conduct some additional tests of the interaction effect (*Accretive Target* with *Regulated*, *Size*, *Dividend*) in regressions to predict the acquirers. Table 8 reports the results. The interaction of firm size with Accretive Targets is positive and significant at the 1% level both with fixed effects (0.37) and without (0.56), and once it is included the main effect of Accretive Targets is no longer positive. The link between accretion opportunities and acquisitions is thus concentrated among large firms, consistent with the survey evidence. We find no reliable differences for dividend payers, and the interaction with Regulated is significant only in the specification without fixed effects. As in the other tables, the number of takeovers in the industry is positive and leverage is negative in every specification.

4.6 Pooling of Interests

We explore an accounting channel through which management attention to earnings interacts with the number of EPS accretive bidders. Before mid-2001, firms that met the required conditions could account for an acquisition using either the purchase method or the pooling-of-interests method. Under the pooling method, acquirers combine the balance sheets of the two merged firms, while the purchase method requires acquirers to record the premium as goodwill and amortize it for a period of years. Although neither method affects cash flow, the pooling method allows acquirers to avoid the purchase method's negative effect on accounting earnings and is generally preferred by acquirers, especially when CEOs have more earnings-based compensation (Aboody et al. 2000)[10]. A firm with more EPS-accretive bidders should therefore be even more likely to become a target when those bidders can use pooling. For business combinations initiated after June 30, 2001, SFAS 141 requires the purchase method. Acquirers no longer amortize goodwill, but they must test it annually for impairment, and impairment charges can reduce the combined firm's earnings.[11] Because pooling was available only for stock deals, and our count of viable bidders also assumes stock financing, the incentive for EPS accretive bidders to acquire should be higher before pooling was eliminated.

Table 9 compares the effect of accretive bidders on takeover likelihood before and after pooling was eliminated. Because the accounting change took effect in mid-2001, we exclude outcome years 2001 and 2002 and compare 1991–2000 with 2003–2012 (columns 1 to 3) and, as a check, 1989–2000 with 2003–2014 (columns 4 to 6). The coefficient on Accretive Bidders is 0.67 in the pre-period and 0.41 in the post-period using ten-year windows, and 0.63 and 0.34 using

[10] Survey evidence in Graham and Harvey (2001) suggests that some managers preferred stock as acquisition consideration, especially when pooling-of-interests accounting was available.
[11] Yehuda et al. (2019) show that acquirers tend to allocate their overpayment to in-process R&D and intangible assets in negative NPV acquisition to avoid future poor performance.

twelve-year windows. In the combined regressions, which include Accretive Bidders together with its interaction with a pre-period indicator, the interaction is 0.24 (significant at the 10% level) and 0.33 (significant at the 1% level), so the effect of accretion opportunities was roughly 60% larger to more than twice as large when pooling was available. These results are consistent with our hypothesis that pooling amplified the incentive to pursue accretive acquisitions. With shorter windows of three and five years on each side, the pre-post difference is not statistically significant. The analogous acquirer-side comparison, which interacts Accretive Targets with the pre-period indicator in the acquirer regression, is not statistically significant (untabulated). Because the two periods also differ in other respects, including the late-1990s valuation boom, Regulation FD (2000) and the Sarbanes-Oxley Act (2002), we view this evidence as consistent with an accounting channel rather than as proof of one.

4.7 Earnings-Based Compensation and Bidder Liquidity

If managers pursue accretive acquisitions because their pay is tied to earnings, accretion opportunities should matter more for firms whose own executives have earnings-sensitive pay. Because an EPS-accretive deal raises the acquirer's reported EPS, it should be more attractive to such executives, so a firm with many accretive targets should be more likely to become an acquirer when its pay is more sensitive to earnings. We obtain our earnings-based pay-performance sensitivity (PPS) measures from Bennett et al. (2017), who use Incentive Lab data on executive compensation contracts to characterize the sensitivity of managerial pay to earnings-based performance targets.[12] These measures are available from 1997. We define PPS incentive as the empirical CDF of the firm's own earnings-based PPS across all firms with available data in that

[12] We thank Ben Bennett for providing the PPS data.

year and add it, together with its interaction with Accretive Targets, to the acquirer regressions of Table 7.[13]

Table 10 reports the results. The interaction between Accretive Targets and PPS incentive is positive in every specification: 0.40 to 0.46 and significant at the 5% level with year and industry fixed effects (columns 1 to 3), and 0.34 to 0.38 without them (columns 4 to 6), significant at the 5% level in column 4 and at the 10% level in columns 5 and 6. The main effects measure the relation of each variable when the other is at the bottom of its distribution, so their negative signs are not informative on their own. Because the interaction coefficient in a probit model is not the interaction effect on the probability (Ai and Norton, 2003), Appendix Table A7 reports the implied probability effects. In column 2, moving Accretive Targets from its 25th to its 75th percentile changes the annual probability of an acquisition by −0.15 percentage points for a firm at the 25th percentile of PPS incentive, by 0.38 percentage points at the median and by 0.92 percentage points at the 75th percentile. The individual effects are imprecisely estimated, but the difference between the effects at the 75th and 25th percentiles of PPS incentive is significant in columns 2 and 3 ($p = 0.037$ and 0.040). The bottom quartile of PPS includes executives with negative earnings-based PPS, for whom an accretive deal offers little or no compensation benefit, so the absence of an effect in that group is consistent with the compensation channel.

Two caveats apply. First, the compensation sample is considerably smaller than our main sample (24,421 firm-years in the fixed-effects specifications, compared with 91,488 in Table 7), begins in 1997 and is tilted toward larger firms with available contract data. In this subsample, the target-side relation between Accretive Bidders and takeover likelihood is positive but no longer

[13] We focus on earnings-based PPS because stock-based measures of pay-performance sensitivity (Jensen and Murphy, 1990; Core and Guay, 2002) are not directly tied to reported EPS.

statistically significant once controls are included (Appendix Table A8). Second, earnings-based pay may be correlated with other firm characteristics, such as governance or growth opportunities, that also shape acquisition behavior. We therefore view the compensation evidence as supportive of, rather than conclusive for, an incentive channel.

Finally, we examine whether bidder liquidity matters. If financing frictions limit bidders' ability to act on accretive opportunities, a firm should be more likely to become a target when more of its accretive bidders are cash-rich (Harford, 1999). Table 11 adds the number of accretive bidders with high cash-to-assets ratios, transformed to its annual CDF, to the baseline target specification, again controlling for the total number of accretive bidders. With fixed effects, the coefficient is positive and significant at the 10% level once controls are included (0.15 and 0.14); without fixed effects, it is negative and significant at the 10% level in one of three specifications. When we instead interact Accretive Targets with the firm's own liquidity in the acquirer regression, the interaction is not significant (untabulated). We therefore find no reliable evidence that bidder liquidity amplifies the effect of accretion opportunities. Taken together, Tables 10 and 11 point to managerial compensation, rather than financing capacity, as one reason accretion opportunities shape who acquires whom.

5 Conclusion

In this paper, we examine whether EPS accretion opportunities help explain which firms merge and with whom. We show that the number of EPS-accretive bidders available to a firm predicts both overall merger activity and the likelihood that the firm becomes a target, and that the number of accretive targets available to a firm predicts whether it becomes an acquirer. Moving from the bottom to the top of the distribution of accretive bidders raises the annual probability of becoming a target from 4.3% to about 6.9%. These results hold after controlling for the valuation-

based predictors emphasized in prior work and are not simply a restatement of industry merger waves.

Four pieces of evidence tie these results to earnings. Counts of viable bidders based on book value have little incremental predictive power once the EPS-based count is included. The effect of accretion opportunities was substantially larger before the 2001 elimination of pooling-of-interests accounting, when acquirers could more easily preserve reported earnings. The link between accretion opportunities and acquisitions is concentrated among large firms, which survey evidence identifies as most concerned with EPS dilution. And among firms with compensation data, it is stronger when the acquirer's executive pay is more sensitive to earnings; we find no reliable evidence that bidder liquidity plays a similar role. Because accretion depends only on the two firms' price-to-forecast-earnings ratios, our counts cannot be fully separated from a firm's relative valuation, and several results are weaker without year and industry fixed effects. The compensation evidence also comes from a smaller sample of mostly larger firms beginning in 1997. We therefore place most weight on the patterns that a pure valuation explanation does not predict: the concave form implied by our model, the concentration of the effect among large and earnings-incentivized acquirers, and its variation around the elimination of pooling.

Taken together, the results suggest that EPS considerations shape not only deal structure and pricing, as documented in prior work, but also whether deals occur and which firms pair with one another. Examining the value consequences of EPS-motivated mergers, and identifying why managers weight reported EPS so heavily, remain productive directions for future research.

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Tables

Table 1
Summary Statistics

Panel A: Firm Characteristics

	Acquirers			Targets			Non-mergers		
	N	Mean	Median	N	Mean	Median	N	Mean	Median
Characteristics									
Industry takeovers	3,992	19.154	13.000	3,959	15.735	9.000	83,596	9.868	5.000
Size (log assets)	3,992	8.014	8.033	3,959	6.345	6.248	83,596	6.803	6.702
Leverage	3,992	0.205	0.172	3,959	0.220	0.183	83,596	0.228	0.194
Industry firms	3,992	396.296	385.000	3,959	337.369	275.000	83,596	278.364	186.000
Price / forecast EPS	3,992	43.025	18.964	3,959	34.168	15.811	83,596	45.125	18.580
Bidder variables									
Accretive bidders	3,992	60.508	38.000	3,959	64.526	40.000	83,596	45.025	23.000
Book Bidders	3,955	97.655	61.000	3,891	121.879	70.000	82,171	93.072	48.000
RIM Bidders	3,205	51.225	29.000	2,575	53.816	31.000	57,098	33.700	17.000
Target variables									
Accretive targets	3,992	68.594	40.000	3,959	46.154	21.000	83,596	47.105	21.000
Book Targets	3,955	185.990	128.000	3,891	127.860	73.000	82,171	110.334	57.000
RIM Targets	3,205	54.498	34.000	2,575	34.256	15.000	57,098	34.978	16.000

Panel B: Differences in Means

	Non-merger – acquirer		Non-merger – target		Target – acquirer	
	t	p	t	p	t	p
Characteristics						
Industry takeovers	-40.019	<0.001	-25.393	<0.001	-7.605	<0.001
Size (log assets)	-35.880	<0.001	13.547	<0.001	-38.105	<0.001
Leverage	7.088	<0.001	2.427	0.015	3.626	<0.001
Industry firms	-29.140	<0.001	-14.528	<0.001	-9.856	<0.001
Price / forecast EPS	0.607	0.544	3.154	0.002	-2.865	0.004
Bidder variables						
Accretive bidders	-16.942	<0.001	-21.148	<0.001	2.727	0.006
Book Bidders	-2.452	0.014	-15.131	<0.001	8.745	<0.001
RIM Bidders	-21.345	<0.001	-22.052	<0.001	1.654	0.098
Target variables						
Accretive targets	-20.977	<0.001	0.935	0.350	-14.562	<0.001
Book Targets	-34.400	<0.001	-8.013	<0.001	-16.131	<0.001
RIM Targets	-22.488	<0.001	0.762	0.446	-14.075	<0.001

Table 1
Summary Statistics (continued)

Panel C: Correlations

	1	2	3	4	5	6	7	8	9	10	11
Characteristics											
1. Industry takeovers	1.00										
2. Size (log assets)	0.02	1.00									
3. Leverage	-0.18	0.21	1.00								
4. Industry firms	0.81	0.02	-0.18	1.00							
5. Price / forecast EPS	-0.00	-0.04	-0.05	0.01	1.00						
Bidder variables											
6. Accretive bidders	0.57	0.06	-0.09	0.65	-0.13	1.00					
7. Book Bidders	0.55	-0.08	-0.11	0.65	-0.04	0.64	1.00				
8. RIM Bidders	0.55	0.05	-0.14	0.58	-0.11	0.77	0.52	1.00			
Target variables											
9. Accretive targets	0.55	0.03	-0.22	0.59	0.15	-0.02	0.23	0.08	1.00		
10. Book Targets	0.64	0.04	-0.18	0.72	0.08	0.33	0.07	0.32	0.62	1.00	
11. RIM Targets	0.57	0.12	-0.21	0.57	0.10	0.14	0.26	0.01	0.80	0.57	1.00

Panel D: Industry-year Merger Rates

	N	Spearman	Pearson
Bidder variables			
Accretive bidders	91,547	0.3748	0.3286
Book Bidders	91,547	0.3700	0.3058
RIM Bidders	90,539	0.4039	0.3762
Target variables			
Accretive targets	91,547	0.3673	0.3456
Book Targets	91,547	0.3653	0.3047
RIM Targets	90,539	0.4114	0.3886

Notes: Panel A reports N, means, and medians for 91,547 distinct firm-years. Acquirers and targets are classified by current-year status; non-mergers are neither. All characteristics and counts are lagged one calendar year. Bidder counts measure eligible buyers of the focal firm; target counts measure its eligible acquisition targets. Counts use distinct firms in the same two-digit SIC industry-year at a 20% premium and exclude self-matches. RIM Bidders and RIM Targets use the three-year RIM. Panel B reports two-sided, equal-variance two-sample t-tests, with the sign equal to the first group minus the second. Panel C reports Pearson correlations on 62,878 complete observations across all eleven variables, in the same order as Panels A and B. Panel D correlates industry-year medians of lagged counts with the industry-year merger rate defined as, takeover counts divided by unique-firm counts in the input universe. Each retained firm-year receives equal weight, so industry-years with more retained firms receive greater weight; N reports available firm-year observations for each measure.

Table 2
Merger Activity and Accretion Opportunities

	(1)	(2)	(3)	(4)
Accretive bidders	0.4349***	0.5408***	0.4347***	0.5319***
	(0.07924)	(0.1443)	(0.1238)	(0.149)
Accretive targets	0.1315**	0.2182**	0.1323	0.2096**
	(0.05894)	(0.09714)	(0.09031)	(0.1019)
Year fixed effects	Yes	Yes	No	No
Industry fixed effects	Yes	No	Yes	No
Observations	91,520	91,547	91,520	91,547
Pseudo R^2	0.052	0.035	0.032	0.014

Notes: This table reports coefficients from probit regressions of merger participation on Accretive bidders and Accretive targets. The dependent variable equals one if the firm is a takeover target or an acquirer and zero otherwise. Column (1) includes separate year and two-digit SIC industry fixed effects; Column (2) includes only year fixed effects; Column (3) includes only industry fixed effects; Column (4) includes neither. Regressors are lagged one calendar year and transformed using their within-year empirical cumulative distribution functions (CDFs). Coefficients are probit index coefficients. Standard errors clustered by two-digit SIC industry are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table 3
Accretive Bidders and Annual Takeover Activity

	(1)	(2)	(3)
Median accretive bidders, current	12.8***		
	(2.358)		
Median accretive bidders, lagged		13.06***	11.73***
		(2.203)	(2.424)
S&P 500 index, current	-0.1167***	-0.1237***	
	(0.02127)	(0.02058)	
Market-to-book interdecile, current	53.22***	61.95***	
	(19.24)	(18)	
S&P 500 index, lagged			-0.1225***
			(0.02043)
Market-to-book interdecile, lagged			58.66***
			(18.68)
Year FE	No	No	No
Industry FE	No	No	No
Observations	43	43	43
R^2	0.584	0.616	0.614

Notes: This table reports OLS estimates of aggregate takeover activity. This table uses one observation per calendar year from 1980–2022. The dependent variable counts unique target firm-years in the reconstructed input universe. Accretive-bidder counts and market-to-book interdecile dispersion are annual medians in the retained sample; S&P 500 index levels are annual means. Current and lagged regressors are identified in the row labels and enter in their original units. Conventional OLS standard errors are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table 4
Accretive Bidders and Industry Takeover Activity

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Industry mean accretive bidders	0.1242***	0.1312***	0.1315***	0.1296***	0.06614***	0.06838***	0.1309***	0.131***	0.06609***	0.06917***
	(0.001841)	(0.003617)	(0.003648)	(0.003619)	(0.00382)	(0.003839)	(0.003651)	(0.003608)	(0.003824)	(0.003863)
Industry firms		-0.00117**	-0.001237**	-0.001202**	-0.0007787*	-0.000796*	-0.001115**	-0.001343***	-0.000709	-0.000807*
		(0.0005235)	(0.00053)	(0.0005248)	(0.0004613)	(0.0004597)	(0.0005262)	(0.0005202)	(0.0004584)	(0.000457)
Industry mean market-to-book			0.001789	0.001061	0.001957	0.001787				
			(0.002207)	(0.002234)	(0.001963)	(0.001956)				
Ind Average Time Series Error							0.1	-0.4919***	-0.03006	-0.2099
							(0.1455)	(0.1602)	(0.1363)	(0.141)
Ind Average long-run VtoB							0.5642***	0.446***	0.3184**	0.2845*
							(0.17)	(0.1721)	(0.1516)	(0.1512)
All takeovers, prior year				0.008259***		0.002867***		0.008817***		0.003154***
				(0.0006992)		(0.0006413)		(0.0007217)		(0.0006652)
Industry takeovers, prior year					0.5113***	0.4883***			0.5092***	0.4841***
					(0.01656)	(0.01729)			(0.01658)	(0.01734)
Year FE	No	No	No	No	No	No	No	No	No	No
Industry FE	No	No	No	No	No	No	No	No	No	No
Observations	2,707	2,707	2,707	2,621	2,621	2,621	2,706	2,620	2,620	2,620
R^2	0.627	0.628	0.628	0.647	0.727	0.729	0.629	0.650	0.728	0.730

Notes: This table reports OLS estimates of aggregate takeover activity using one observation per two-digit SIC industry and calendar year. The dependent variable is the count of merger announcements in industry j, year t. Columns (7)–(10) correspond to Columns (3)–(6), respectively, replacing industry mean market-to- book with industry means of Time Series Error2 and Long Run VtoB2. Accretive bidders, firm counts, market-to-book and the two valuation components are industry-year means of available lagged firm-level values. Prior takeover activity uses calendar-year lags of counts. No year or industry fixed effects are included. Conventional OLS standard errors are reported in parentheses. *p < 0.10, **p < 0.05, and ***p < 0.01.

Table 5
Takeover Likelihood and the Number of Accretive Bidders

	(1)	(2)	(3)	(4)	(5)	(6)
Accretive Bidders	0.7100***	0.2733***	0.2867**	0.5015***	0.2598**	0.2277*
	(0.0909)	(0.1054)	(0.1140)	(0.0760)	(0.1304)	(0.1340)
Ind P/FE SD		-0.0596			-0.0872	
		(0.0478)			(0.0888)	
Ind P/FE Interdecile			-0.0708			-0.0319
			(0.0682)			(0.0804)
Ind Num Firms		-0.1513	-0.1723		-0.1194	-0.1393
		(0.1265)	(0.1313)		(0.1579)	(0.1558)
P/FE		-0.3398***	-0.3291***		-0.3296***	-0.3534***
		(0.0804)	(0.0826)		(0.0812)	(0.0870)
Size		0.3162**	0.3156**		0.2888*	0.2997*
		(0.1369)	(0.1368)		(0.1506)	(0.1533)
Size Square		-0.6785***	-0.6779***		-0.5888***	-0.5883***
		(0.1214)	(0.1215)		(0.1334)	(0.1356)
Leverage		0.0904**	0.0909**		0.0702	0.0686
		(0.0455)	(0.0453)		(0.0546)	(0.0540)
Ind Num Takeover		0.3592***	0.3562***		0.3365***	0.3541***
		(0.0806)	(0.0828)		(0.1206)	(0.1180)
Year FE	Yes	Yes	Yes	No	No	No
Industry FE	Yes	Yes	Yes	No	No	No
Observations	91,520	91,520	91,520	91,547	91,547	91,547
Pseudo R-squared	0.0466	0.0542	0.0542	0.0113	0.0194	0.0192

Notes: This table reports coefficients from probit regressions of takeover likelihood and the number of viable bidders. The dependent variable equals one if the firm is a takeover target and zero otherwise. Columns (1)–(3) include separate year and industry fixed effects; Columns (4)–(6) exclude them. Peers are defined by two-digit SIC industry, and the acquisition premium is 20%. Columns (2) and (5) include Ind P/FE SD; Columns (3) and (6) include Ind P/FE Interdecile. Covariates are lagged one calendar year. Continuous regressors use annual CDFs. The sample uses available observations from 1980-2022. Standard errors clustered by two-digit SIC industry are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table 6
Takeover Likelihood and the Number of Accretive, Book, and RIM Bidders

	(1)	(2)	(3)	(4)	(5)	(6)
Accretive Bidders	0.9135***	0.3518**	0.3751**	0.6369***	0.1591	0.1044
	(0.1156)	(0.1640)	(0.1668)	(0.1110)	(0.1400)	(0.1470)
Book Bidders	0.1888	0.1899	0.1822	-0.2647**	0.1855**	0.1370
	(0.1611)	(0.1284)	(0.1253)	(0.1031)	(0.0936)	(0.0895)
RIM Bidders	0.3151***	0.2093***	0.2071***	0.1999**	0.1855**	0.2004***
	(0.0924)	(0.0662)	(0.0670)	(0.1010)	(0.0762)	(0.0748)
Ind P/FE SD		-0.1235**			-0.1123	
		(0.0607)			(0.1014)	
Ind P/FE Interdecile			-0.1255			-0.0111
			(0.0827)			(0.0852)
Ind Num Firms		-0.1794	-0.2217		-0.2267*	-0.2325*
		(0.1835)	(0.1872)		(0.1212)	(0.1325)
P/FE		-0.5941***	-0.5770***		-0.6728***	-0.7054***
		(0.1279)	(0.1273)		(0.1110)	(0.1155)
Firm Specific Error2	0.1773**	0.3678***	0.3626***	-0.0848*	0.3848***	0.3623***
	(0.0775)	(0.0879)	(0.0870)	(0.0508)	(0.0742)	(0.0758)
Time Series Error2	-0.0331	0.0765	0.0765	-0.0549	0.0852**	0.0775*
	(0.0602)	(0.0466)	(0.0473)	(0.0467)	(0.0413)	(0.0404)
Long Run VtoB2	-0.0114	-0.1077**	-0.1118**	-0.1668***	-0.1462***	-0.1768***
	(0.0810)	(0.0526)	(0.0508)	(0.0468)	(0.0438)	(0.0453)
Size		0.4307***	0.4294***		0.4252***	0.4408***
		(0.1541)	(0.1545)		(0.1546)	(0.1565)
Size Square		-0.8339***	-0.8330***		-0.7780***	-0.7813***
		(0.1381)	(0.1386)		(0.1415)	(0.1435)
Leverage		0.0565	0.0575		0.1074*	0.1063*
		(0.0594)	(0.0593)		(0.0646)	(0.0638)
Ind Num Takeover		0.3233***	0.3225***		0.3334***	0.3684***
		(0.1095)	(0.1108)		(0.1169)	(0.1179)
Year FE	Yes	Yes	Yes	No	No	No
Industry FE	Yes	Yes	Yes	No	No	No
Observations	62,634	62,634	62,634	62,672	62,672	62,672
Pseudo R-squared	0.0592	0.0694	0.0694	0.0206	0.0346	0.0342

Notes: This table reports coefficients from probit regressions of takeover likelihood and the number of viable bidders. The dependent variable equals one if the firm is a takeover target and zero otherwise. Columns (1)–(3) include separate year and industry fixed effects; Columns (4)–(6) exclude them. Peers are defined by two-digit SIC industry, and the acquisition premium is 20%. Columns (2) and (5) include Ind P/FE SD; Columns (3) and (6) include Ind P/FE Interdecile. Covariates are lagged one calendar year. Continuous regressors use annual CDFs. RIM uses three-year residual income model. The sample uses available observations from 1980-2022. Standard errors clustered by two-digit SIC industry are reported in parentheses. *p < 0.10, **p < 0.05, and ***p < 0.01.

Table 7
Acquisition Likelihood and the Number of Accretive Targets

	(1)	(2)	(3)	(4)	(5)	(6)
Accretive Targets	0.2361***	0.2312**	0.2384**	0.5763***	0.0415	0.0464
	(0.0456)	(0.0999)	(0.0980)	(0.1395)	(0.1221)	(0.1302)
P/FE		-0.0133	-0.0204		0.1524	0.1504
		(0.1004)	(0.0982)		(0.1203)	(0.1293)
Ind Num Firms		-0.1995	-0.1834		-0.0150	0.0082
		(0.1809)	(0.1804)		(0.1328)	(0.1436)
Ind P/FE SD			-0.0643			-0.0869
			(0.0447)			(0.0917)
Ind P/FE Interdecile		-0.0615			-0.0582	
		(0.0590)			(0.0783)	
Size		1.0784***	1.0786***		1.0544***	1.0503***
		(0.2222)	(0.2235)		(0.2024)	(0.1983)
Size Square		0.2150	0.2145		0.1583	0.1569
		(0.1849)	(0.1864)		(0.1724)	(0.1724)
Leverage		-0.2073***	-0.2070***		-0.1658*	-0.1674*
		(0.0503)	(0.0499)		(0.0880)	(0.0874)
Ind Num Takeover		0.7249***	0.7247***		0.7579***	0.7501***
		(0.1031)	(0.1018)		(0.1229)	(0.1223)
Year FE	Yes	Yes	Yes	No	No	No
Industry FE	Yes	Yes	Yes	No	No	No
Observations	91,488	91,488	91,488	91,547	91,547	91,547
Pseudo R-squared	0.0591	0.1119	0.1119	0.0146	0.0892	0.0893

Notes: This table reports coefficients from probit regressions of acquisition likelihood and the number of accretive targets. The dependent variable equals one if the firm is an acquirer and zero otherwise. Columns (1)–(3) include separate year and industry fixed effects; Columns (4)–(6) exclude them. Peers are defined by two-digit SIC industry, and the acquisition premium is 20%. Columns (2) and (5) include Ind P/FE Interdecile; Columns (3) and (6) include Ind P/FE SD. Covariates are lagged one calendar year. Continuous regressors use annual CDFs. The sample uses available observations from 1980-2022. Standard errors clustered by two-digit SIC industry are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table 8
Acquirer Characteristics and Accretion Opportunities

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Size × accretive targets		0.3725***				0.5561***		
		(0.09263)				(0.09886)		
Dividend × accretive targets			-0.02536				0.07436	
			(0.09611)				(0.1076)	
Regulated × accretive targets				0.01716				0.2769**
				(0.06731)				(0.1175)
Accretive targets	0.2138***	-0.01904	0.2334***	0.2063***	0.1915***	-0.1543**	0.1482*	0.08349
	(0.03544)	(0.06323)	(0.07777)	(0.04499)	(0.02543)	(0.07458)	(0.08326)	(0.07057)
Dividend	-0.05602		-0.0419		-0.08939*		-0.138***	
	(0.04876)		(0.03346)		(0.04881)		(0.03939)	
Regulated	-0.1671***			-0.1691***	-0.06721			-0.2403**
	(0.01057)			(0.03659)	(0.06511)			(0.1144)
Industry P/FE interdecile	-0.05996	-0.06503	-0.06065	-0.06392	-0.09127	0.000427	-0.03807	-0.04433
	(0.06004)	(0.0601)	(0.05996)	(0.06047)	(0.08838)	(0.09127)	(0.08484)	(0.09583)
Industry firms	-0.2059	-0.1916	-0.2066	-0.1953	-0.0336	-0.08955	-0.08497	-0.0103
	(0.1893)	(0.1915)	(0.1893)	(0.1915)	(0.1135)	(0.1342)	(0.1226)	(0.1275)
Size	1.09***	0.8624***	1.098***	1.072***	1.116***	0.7046***	1.098***	1.026***
	(0.2109)	(0.2215)	(0.2093)	(0.2235)	(0.1863)	(0.2034)	(0.186)	(0.2066)
Size squared	0.2407	0.2178	0.2316	0.2229	0.1874	0.1878	0.1852	0.2078
	(0.1851)	(0.1867)	(0.182)	(0.1872)	(0.1713)	(0.1746)	(0.1663)	(0.1764)
Leverage	-0.2087***	-0.2119***	-0.2069***	-0.2082***	-0.1661*	-0.1623*	-0.1599*	-0.1735**
	(0.04642)	(0.04938)	(0.04709)	(0.04934)	(0.08509)	(0.09135)	(0.09259)	(0.08143)
Industry takeovers	0.7311***	0.7295***	0.7306***	0.726***	0.7274***	0.7395***	0.7362***	0.7259***
	(0.1017)	(0.1017)	(0.102)	(0.102)	(0.1129)	(0.1187)	(0.1209)	(0.1092)
Year fixed effects	Yes	Yes	Yes	Yes	No	No	No	No
Industry fixed effects	Yes	Yes	Yes	Yes	No	No	No	No
Observations	91,094	91,488	91,094	91,488	91,152	91,547	91,152	91,547
Pseudo R^2	0.113	0.112	0.113	0.112	0.090	0.090	0.090	0.090

Notes: This table reports coefficients from probit regressions of acquisition likelihood on accretion opportunities and acquirer characteristics. The dependent variable equals one if the firm is an acquirer and zero otherwise. Columns (1)–(4) include separate year and industry fixed effects; Columns (5)–(8) exclude them. Covariates are lagged one calendar year. Continuous regressors use annual CDFs. Interactions are products of the displayed CDF variables or indicators. The sample uses available observations from 1980-2022. Standard errors clustered by two-digit SIC industry are reported in parentheses. *p < 0.10, **p < 0.05, and ***p < 0.01.

Table 9
Pooling-of-Interests Accounting and Takeover Likelihood

	10 years per side			12 years per side		
	(1)	(2)	(3)	(4)	(5)	(6)
Sample	Pre	Post	Combined	Pre	Post	Combined
Pre-period	1991–2000		1991–2000	1989–2000		1989–2000
Post-period		2003–2012	2003–2012		2003–2014	2003–2014
Pre-period × accretive bidders			0.2410*			0.3280***
			(0.1387)			(0.1252)
Accretive bidders	0.6652**	0.4094**	0.3841***	0.6306**	0.3357*	0.2420**
	(0.2611)	(0.1741)	(0.1213)	(0.2455)	(0.1937)	(0.1134)
Industry firms	0.4341	-1.0432	-0.4108	0.2007	-0.7469	-0.3502
	(0.3964)	(1.1884)	(0.3230)	(0.3301)	(1.1638)	(0.2941)
Price / forecast EPS	-0.4456***	-0.0222	-0.2773***	-0.4776***	0.0257	-0.2861***
	(0.0913)	(0.1728)	(0.1055)	(0.0861)	(0.1847)	(0.1048)
Ind P/FE SD	-0.1877***	-0.0527	-0.0991	-0.1436**	-0.0777	-0.0824
	(0.0664)	(0.1179)	(0.0682)	(0.0622)	(0.1063)	(0.0624)
Size	0.4163*	-0.0869	0.2134	0.3895*	0.0564	0.2443*
	(0.2287)	(0.2546)	(0.1497)	(0.2126)	(0.2278)	(0.1482)
Size squared	-0.7096***	-0.3362	-0.5853***	-0.6674***	-0.4968**	-0.6233***
	(0.2257)	(0.2335)	(0.1298)	(0.2197)	(0.1943)	(0.1346)
Leverage	0.0054	0.1080	0.0460	0.0296	0.1454	0.0796
	(0.0778)	(0.0985)	(0.0656)	(0.0748)	(0.0922)	(0.0656)
Ind Num takeover	0.1260	-0.2260	0.2018	0.2798**	0.0759	0.3166***
	(0.1620)	(0.2234)	(0.1305)	(0.1343)	(0.2028)	(0.1097)
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	23,476	23,299	47,328	26,656	28,350	55,460
Pseudo R^2	0.0683	0.0364	0.0617	0.0710	0.0321	0.0586

Notes: This table reports coefficients from probit regressions of takeover likelihood before and after the elimination of pooling-of-interests accounting. The dependent variable equals one if the firm is a takeover target and zero otherwise. All columns include separate year and industry fixed effects. Columns (1)–(3) use 1991–2000 and 2003– 2012; Columns (4)–(6) use 1989–2000 and 2003–2014. Outcome years 2001 and 2002 are excluded. All columns include Ind Num takeover and Ind P/FE SD. In combined Columns (3) and (6), Accretive bidders is the post-period coefficient; adding the pre-period interaction gives the pre-period coefficient. Covariates are lagged one calendar year. Continuous regressors use annual CDFs. Standard errors clustered by two-digit SIC industry are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table 10
PPS Incentives and Acquisition Likelihood

	Own PPS5 CDF					
	(1)	(2)	(3)	(4)	(5)	(6)
Accretive Targets × PPS incentive	0.4049**	0.4572**	0.4509**	0.3444**	0.3760*	0.3708*
	(0.1829)	(0.2197)	(0.2201)	(0.1757)	(0.1991)	(0.2051)
Accretive Targets	-0.3008**	-0.1474	-0.1463	-0.2740**	-0.1840	-0.1948
	(0.1451)	(0.1545)	(0.1540)	(0.1317)	(0.1400)	(0.1417)
PPS incentive	-0.2778**	-0.2032	-0.1987	-0.2476**	-0.2041*	-0.1988
	(0.1159)	(0.1268)	(0.1267)	(0.1082)	(0.1198)	(0.1211)
Industry P/FE SD		-0.0003***			-0.0003**	
		(0.0001)			(0.0001)	
Industry P/FE interdecile			-0.0009**			0.0001
			(0.0004)			(0.0003)
Industry firms		0.0003**	0.0003**		0.0004	0.0003
		(0.0002)	(0.0001)		(0.0003)	(0.0003)
Price / forecast EPS		0.0001***	0.0001***		0.0001***	0.0001***
		(0.0000)	(0.0000)		(0.0000)	(0.0000)
Size		0.4503***	0.4490***		0.3337***	0.3394***
		(0.1602)	(0.1600)		(0.1155)	(0.1192)
Size squared		-0.0134	-0.0133		-0.0097	-0.0099
		(0.0099)	(0.0099)		(0.0072)	(0.0073)
Leverage		-0.2976**	-0.2966**		-0.3585**	-0.3678**
		(0.1449)	(0.1456)		(0.1805)	(0.1758)
Ind Num Takeover	1.1528***	1.1329***	1.1422***	0.8859***	0.5314**	0.5554***
	(0.1763)	(0.1900)	(0.1926)	(0.1609)	(0.2182)	(0.2047)
Year fixed effects	Yes	Yes	Yes	No	No	No
Industry fixed effects	Yes	Yes	Yes	No	No	No
Observations	24,421	24,421	24,421	24,742	24,742	24,742
Pseudo R^2	0.0737	0.1230	0.1225	0.0282	0.0704	0.0695

Notes: This table reports probit index coefficients. The dependent variable equals one if the firm is an acquirer and zero otherwise. Columns (1)–(3) include separate year and industry fixed effects; Columns (4)–(6) exclude them. PPS incentive is the empirical CDF of the focal firm's own PPS5 within the source year, across all distinct firms with observed PPS5 before EPS eligibility restrictions. Equal ties receive the same rank; negative and zero PPS are retained. Accretive Targets is the annual CDF of the number of distinct same-industry firms the focal firm could acquire accretively at a 20% premium. Both main effects and their interaction are included. Covariates are lagged one calendar year. Ind Num Takeover retains its annual CDF; other controls retain their original units. Columns (2) and (5) include Industry P/FE SD; Columns (3) and (6) include Industry P/FE interdecile. The sample uses available focal PPS5 observations from 1997–2022. Standard errors clustered by two-digit SIC industry are reported in parentheses. *p < 0.10, **p < 0.05, and ***p < 0.01.

Table 11
Bidder Liquidity and Takeover Likelihood

	(1)	(2)	(3)	(4)	(5)	(6)
Liquid accretive bidders	0.1003	0.1525*	0.1423*	-0.0759	-0.1751	-0.2144*
	(0.0821)	(0.0810)	(0.0797)	(0.1240)	(0.1191)	(0.1262)
Accretive Bidders	0.6572***	0.6471***	0.6457***	0.5480***	0.6131***	0.6253***
	(0.0990)	(0.0953)	(0.0969)	(0.1274)	(0.1224)	(0.1246)
Ind Num Firms		0.0001	0.0000		0.0001	0.0001
		(0.0001)	(0.0001)		(0.0001)	(0.0001)
P/FE		0.0001***	0.0001**		0.0000	-0.0000
		(0.0000)	(0.0000)		(0.0000)	(0.0000)
Ind P/FE SD		-0.0002***			-0.0002*	
		(0.0000)			(0.0001)	
Ind P/FE Interdecile			-0.0005***			-0.0003
			(0.0002)			(0.0002)
Size		0.0572**	0.0563**		0.0289	0.0264
		(0.0276)	(0.0279)		(0.0361)	(0.0364)
Size Square		-0.0085***	-0.0084***		-0.0068***	-0.0067**
		(0.0021)	(0.0021)		(0.0026)	(0.0026)
Leverage		0.1557***	0.1565***		0.1172	0.1130
		(0.0598)	(0.0596)		(0.0842)	(0.0843)
Year FE	Yes	Yes	Yes	No	No	No
Industry FE	Yes	Yes	Yes	No	No	No
Observations	91,520	91,520	91,520	91,547	91,547	91,547
Pseudo R-squared	0.0466	0.0531	0.0532	0.0114	0.0215	0.0210

Notes: This table reports coefficients from probit regressions of takeover likelihood and bidder liquidity. The dependent variable equals one if the firm is a takeover target and zero otherwise. Columns (1)–(3) include separate year and industry fixed effects; Columns (4)–(6) exclude them. Peers are defined by two-digit SIC industry, and the acquisition premium is 20%. Liquid accretive bidders is the number of EPS-accretive bidders whose cash-to-assets ratio is above the 75th percentile across all firms with observed cash-to-assets ratios in the source year. Columns (2) and (5) include Ind P/FE SD; Columns (3) and (6) include Ind P/FE Interdecile. Covariates are lagged one calendar year. Liquid accretive bidders enters as its annual empirical CDF, calculated on the full baseline sample before regression exclusions; ties receive equal CDFs. Accretive Bidders uses its annual CDF; other controls retain their original units. The sample uses available observations from 1980-2022. Standard errors clustered by two-digit SIC industry are reported in parentheses. *p < 0.10, **p < 0.05, and ***p < 0.01.

Figures

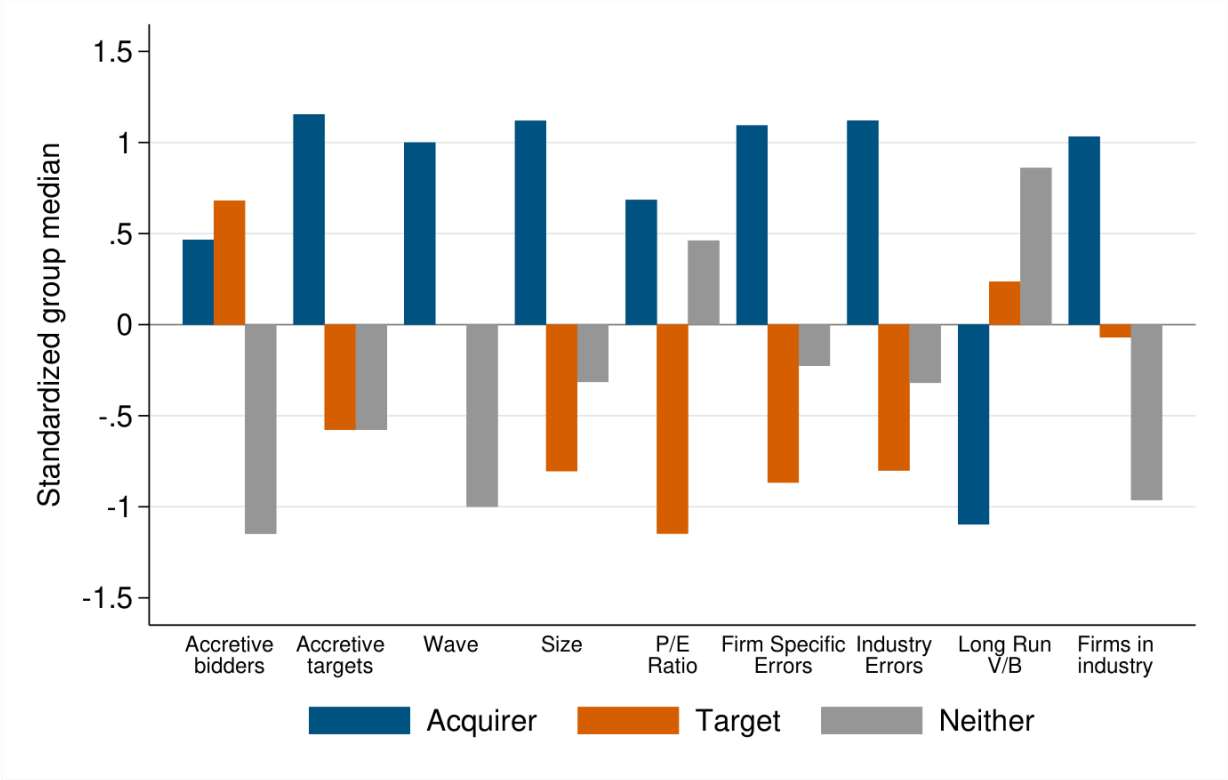
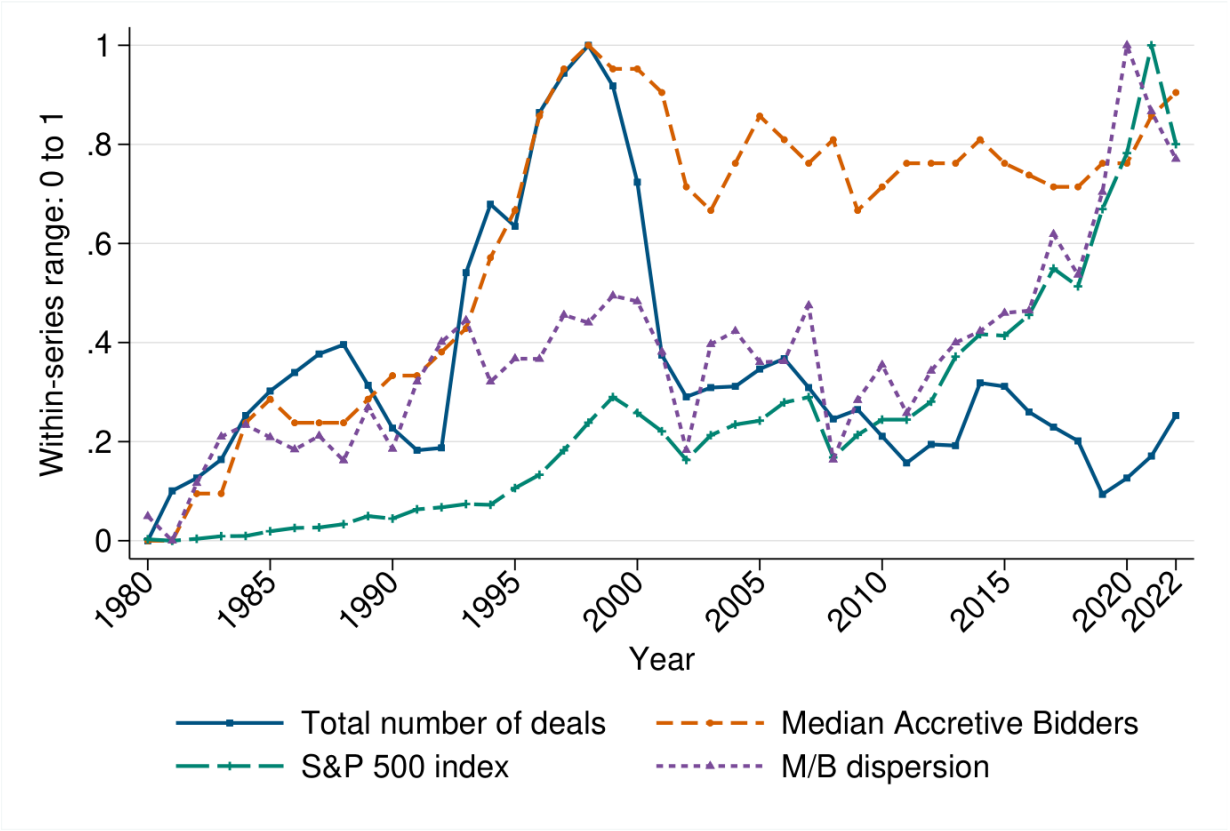


Figure 1. Summary of Empirical Distribution
Each group of three bars shows, from left to right, the acquirer sample, the target sample and the sample of firms that are neither acquirers nor targets. For each variable, we compute the mean and standard deviation of the three sample medians; each bar equals the sample median minus that mean, divided by that standard deviation. Accretive Bidders (Targets) is the number of EPS-based viable buyers (targets) for a firm. Wave is the number of takeovers in the firm's two-digit SIC industry. Size is the natural logarithm of the book value of total assets. P/E ratio is the ratio of stock price to forecast earnings per share. Firm Specific Errors, Industry Errors and Long Run V/B are the firm-specific, time-series (industry) and long-run value-to-book components of the Rhodes-Kropf et al. (2005) decomposition. Firms in industry is the number of firms in the firm's two-digit SIC industry each year.



Year	1980	1981	1982	1983	1984	1985	1986	1987	1988	1989	1990
Total_Number_Deals	37	80	91	107	145	166	182	198	206	171	134
Median_Accretive_Bidders	11	11	13	13	16	17	16	16	16	17	18
S&P_Indx	136	123	141	165	167	211	242	247	278	353	330
M/B_Dispersion	2.62	2.30	3.04	3.64	3.79	3.63	3.48	3.65	3.33	4.02	3.48

Year	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000	2001
Total_Number_Deals	115	117	268	327	308	406	440	464	429	346	197
Median_Accretive_Bidders	18	19	20	23	25	29	31	32	31	31	30
S&P_Indx	417	436	466	459	616	741	970	1229	1469	1320	1148
M/B_Dispersion	4.34	4.85	5.13	4.34	4.64	4.64	5.20	5.10	5.45	5.38	4.73

Year	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012
Total_Number_Deals	161	169	170	185	194	169	142	150	127	104	120
Median_Accretive_Bidders	26	25	27	29	28	27	28	25	26	27	27
S&P_Indx	880	1112	1212	1248	1418	1468	903	1115	1258	1258	1426
M/B_Dispersion	3.47	4.83	4.99	4.59	4.61	5.32	3.34	4.11	4.56	3.94	4.48

Year	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022
Total_Number_Deals	119	173	170	148	135	123	77	91	110	145
Median_Accretive_Bidders	27	28	27	26.5	26	26	27	27	29	30
S&P_Indx	1848	2059	2044	2239	2674	2507	3231	3756	4766	3840
M/B_Dispersion	4.85	4.99	5.23	5.26	6.24	5.72	6.78	8.66	7.81	7.20

Figure 2. Key Variables Across Time

The sample is the sample used in column 4 of Table 2, covering 1980–2022. Total_Number_Deals is the number of deals involving public firms announced each year. S&P_Indx is the level of the S&P 500 index at the end of the calendar year. M/B_Dispersion is the annual median of the industry interdecile range of the market-to-book ratio, and Median_Accretive_Bidders is the annual median of the lagged number of accretive bidders. Each series is scaled by its own range. The data are tabulated below the graph.

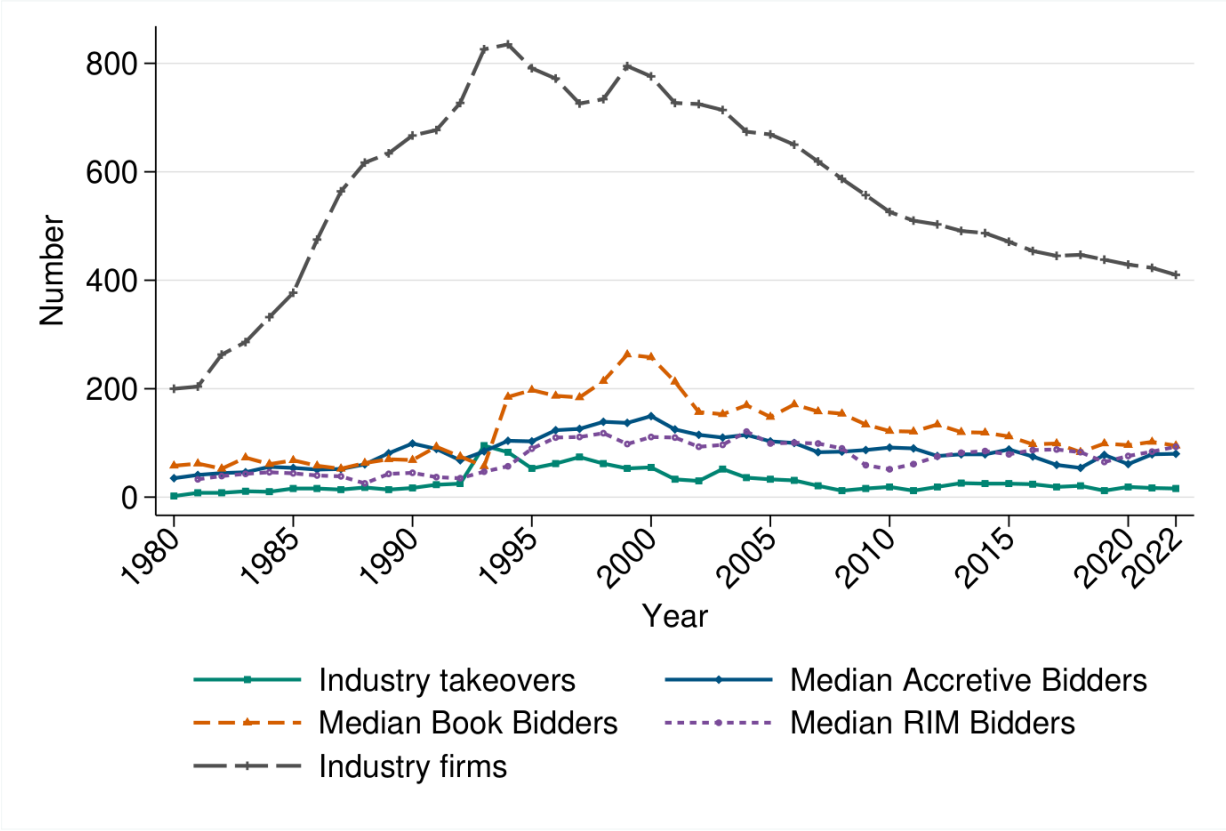


Figure 3. Actual Takeovers among Depository Institutions

The sample is the sample used in column 4 of Table 2, covering 1980–2022, restricted to depository institutions (two-digit SIC 60). Total ind takeover is the number of deals announced in the industry each year. The figure also shows the number of firms in the industry (total num firms) and the annual medians of the lagged numbers of Accretive Bidders (median feps bidders), Book Bidders (median book bidders) and RIM Bidders (median value bidders).

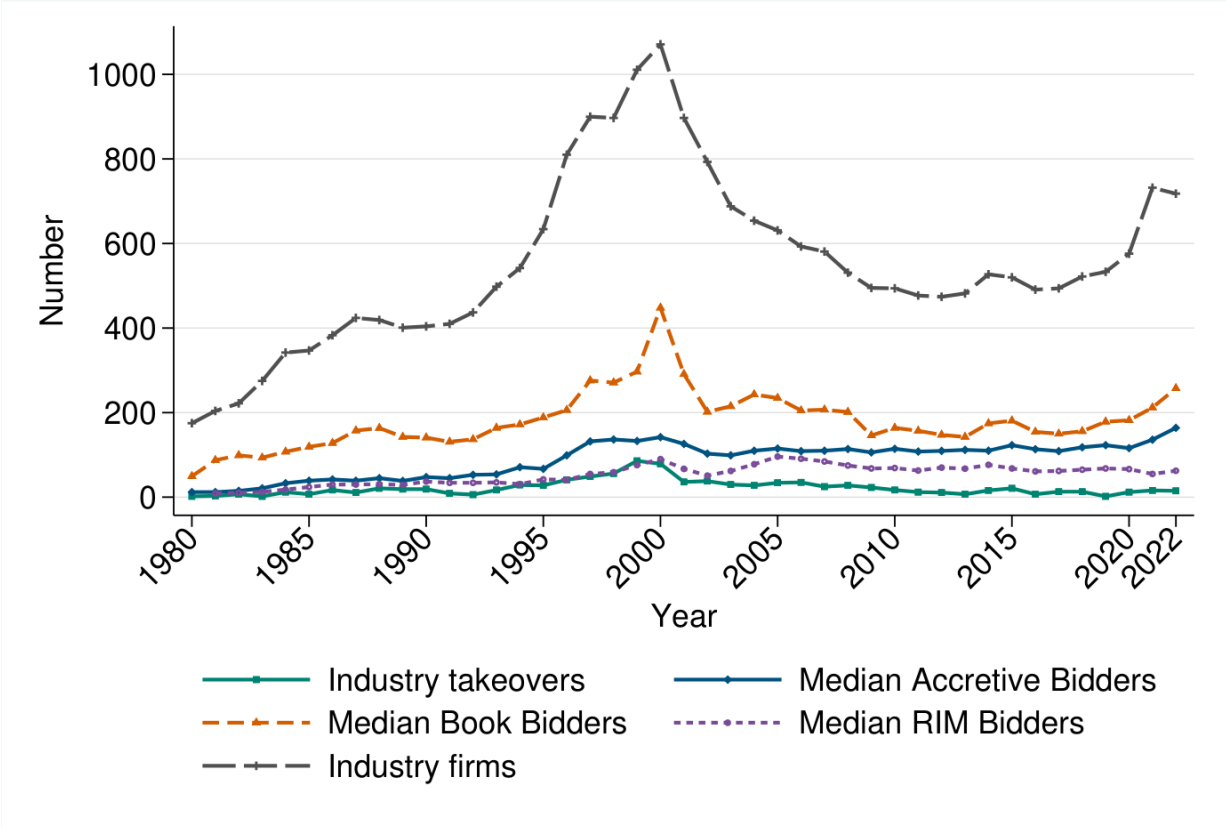


Figure 4. Actual Takeovers among Business Services Firms

The sample is the sample used in column 4 of Table 2, covering 1980–2022, restricted to business services firms (two-digit SIC 73). Total ind takeover is the number of deals announced in the industry each year. The figure also shows the number of firms in the industry (total num firms) and the annual medians of the lagged numbers of Accretive Bidders (median feps bidders), Book Bidders (median book bidders) and RIM Bidders (median value bidders).

Appendix A: Additional Tables

Table A1
Takeover Likelihood and the Logarithm of Accretive Bidders

	(1)	(2)	(3)	(4)	(5)	(6)
log(Accretive Bidders)	0.1421***	0.0542***	0.0571***	0.1074***	0.0926***	0.0874***
	(0.0189)	(0.0175)	(0.0184)	(0.0167)	(0.0205)	(0.0213)
Ind P/FE SD		-0.0667			-0.1258	
		(0.0471)			(0.0816)	
Ind P/FE Interdecile			-0.0779			-0.0660
			(0.0654)			(0.0753)
Ind Num Firms		-0.1400	-0.1632		-0.2354	-0.2666*
		(0.1291)	(0.1334)		(0.1518)	(0.1446)
P/FE		-0.3367***	-0.3255***		-0.2197***	-0.2382***
		(0.0730)	(0.0713)		(0.0648)	(0.0686)
Size		0.3624***	0.3616***		0.3295**	0.3397**
		(0.1376)	(0.1375)		(0.1479)	(0.1506)
Size Square		-0.7122***	-0.7114***		-0.6264***	-0.6248***
		(0.1225)	(0.1224)		(0.1285)	(0.1310)
Leverage		0.0860*	0.0867*		0.0714	0.0714
		(0.0444)	(0.0443)		(0.0541)	(0.0539)
Ind Num Takeover		0.3554***	0.3523***		0.3317***	0.3480***
		(0.0826)	(0.0840)		(0.1183)	(0.1165)
Year FE	Yes	Yes	Yes	No	No	No
Industry FE	Yes	Yes	Yes	No	No	No
Observations	88,148	88,148	88,148	88,157	88,157	88,157
Pseudo R-squared	0.0455	0.0529	0.0530	0.0115	0.0194	0.0191

Notes: This table reports coefficients from probit regressions of takeover likelihood and the number of viable bidders. The dependent variable equals one if the firm is a takeover target and zero otherwise. Columns (1)–(3) include separate year and industry fixed effects; Columns (4)–(6) exclude them. Peers are defined by two-digit SIC industry, and the acquisition premium is 20%. Columns (2) and (5) include Ind P/FE SD; Columns (3) and (6) include Ind P/FE Interdecile. Covariates are lagged one calendar year. Continuous regressors use annual CDFs except the stated logarithmic transformation. The natural logarithm is applied to the raw Accretive Bidders count; zero counts are excluded. The sample uses available observations from 1980-2022. Standard errors clustered by two-digit SIC industry are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table A2
Takeover Likelihood and the Logarithm of One Plus Accretive Bidders

	(1)	(2)	(3)	(4)	(5)	(6)
log(1 + Accretive Bidders)	0.1482***	0.0514**	0.0539**	0.1152***	0.0979***	0.0931***
	(0.0196)	(0.0216)	(0.0229)	(0.0165)	(0.0230)	(0.0237)
Ind P/FE SD		-0.0569			-0.1160	
		(0.0465)			(0.0831)	
Ind P/FE Interdecile			-0.0659			-0.0628
			(0.0652)			(0.0758)
Ind Num Firms		-0.1376	-0.1572		-0.2356	-0.2649*
		(0.1238)	(0.1284)		(0.1480)	(0.1410)
P/FE		-0.3567***	-0.3473***		-0.2187***	-0.2345***
		(0.0759)	(0.0762)		(0.0663)	(0.0704)
Size		0.3157**	0.3150**		0.2852*	0.2942*
		(0.1360)	(0.1359)		(0.1474)	(0.1501)
Size Square		-0.6773***	-0.6767***		-0.5874***	-0.5859***
		(0.1208)	(0.1208)		(0.1295)	(0.1318)
Leverage		0.0896**	0.0901**		0.0770	0.0771
		(0.0456)	(0.0455)		(0.0552)	(0.0549)
Ind Num Takeover		0.3615***	0.3591***		0.3326***	0.3470***
		(0.0804)	(0.0825)		(0.1155)	(0.1135)
Year FE	Yes	Yes	Yes	No	No	No
Industry FE	Yes	Yes	Yes	No	No	No
Observations	91,520	91,520	91,520	91,547	91,547	91,547
Pseudo R-squared	0.0464	0.0541	0.0541	0.0132	0.0209	0.0206

Notes: This table reports coefficients from probit regressions of takeover likelihood and the number of viable bidders. The dependent variable equals one if the firm is a takeover target and zero otherwise. Columns (1)–(3) include separate year and industry fixed effects; Columns (4)–(6) exclude them. Peers are defined by two-digit SIC industry, and the acquisition premium is 20%. Columns (2) and (5) include Ind P/FE SD; Columns (3) and (6) include Ind P/FE Interdecile. Covariates are lagged one calendar year. Continuous regressors use annual CDFs except the stated logarithmic transformation. The natural logarithm is applied to one plus the raw Accretive Bidders count; zero counts are retained. The sample uses available observations from 1980-2022. Standard errors clustered by two-digit SIC industry are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table A3
Acquisition Likelihood and the Logarithm of Accretive Targets

	(1)	(2)	(3)	(4)	(5)	(6)
log(Accretive Targets)	0.0430***	0.0534***	0.0540***	0.1146***	0.0559**	0.0518**
	(0.0109)	(0.0138)	(0.0137)	(0.0254)	(0.0226)	(0.0230)
P/FE		-0.0587	-0.0616		-0.0222	0.0021
		(0.0743)	(0.0730)		(0.1107)	(0.1191)
Ind Num Firms		-0.2191	-0.2017		-0.1680	-0.1265
		(0.1911)	(0.1908)		(0.1484)	(0.1581)
Ind P/FE SD			-0.0645			-0.0625
			(0.0462)			(0.0963)
Ind P/FE Interdecile		-0.0478			-0.0065	
		(0.0603)			(0.0906)	
Size		1.0692***	1.0704***		1.0450***	1.0315***
		(0.2182)	(0.2195)		(0.1974)	(0.1932)
Size Square		0.2332	0.2316		0.1800	0.1811
		(0.1813)	(0.1828)		(0.1681)	(0.1681)
Leverage		-0.2222***	-0.2216***		-0.1585*	-0.1579*
		(0.0520)	(0.0516)		(0.0945)	(0.0951)
Ind Num Takeover		0.7329***	0.7294***		0.7644***	0.7499***
		(0.1083)	(0.1060)		(0.1244)	(0.1232)
Year FE	Yes	Yes	Yes	No	No	No
Industry FE	Yes	Yes	Yes	No	No	No
Observations	87,070	87,070	87,070	87,097	87,097	87,097
Pseudo R-squared	0.0579	0.1113	0.1114	0.0143	0.0892	0.0894

Notes: This table reports coefficients from probit regressions of acquisition likelihood and the number of accretive targets. The dependent variable equals one if the firm is an acquirer and zero otherwise. Columns (1)–(3) include separate year and industry fixed effects; Columns (4)–(6) exclude them. Peers are defined by two-digit SIC industry, and the acquisition premium is 20%. Columns (2) and (5) include Ind P/FE Interdecile; Columns (3) and (6) include Ind P/FE SD. Covariates are lagged one calendar year. Continuous regressors use annual CDFs except the stated logarithmic transformation. The natural logarithm is applied to the raw Accretive Targets count; zero counts are excluded. The sample uses available observations from 1980-2022. Standard errors clustered by two-digit SIC industry are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table A4
Acquisition Likelihood and the Logarithm of One Plus Accretive Targets

	(1)	(2)	(3)	(4)	(5)	(6)
log(1 + Accretive Targets)	0.0493***	0.0506***	0.0519***	0.1218***	0.0581**	0.0550**
	(0.0102)	(0.0174)	(0.0171)	(0.0259)	(0.0228)	(0.0237)
P/FE		-0.0275	-0.0339		-0.0135	0.0048
		(0.0892)	(0.0875)		(0.1120)	(0.1204)
Ind Num Firms		-0.2014	-0.1854		-0.1393	-0.1050
		(0.1818)	(0.1812)		(0.1355)	(0.1463)
Ind P/FE SD			-0.0632			-0.0593
			(0.0450)			(0.0927)
Ind P/FE Interdecile		-0.0589			-0.0130	
		(0.0590)			(0.0847)	
Size		1.0746***	1.0748***		1.0365***	1.0260***
		(0.2232)	(0.2245)		(0.2014)	(0.1976)
Size Square		0.2181	0.2175		0.1719	0.1723
		(0.1856)	(0.1871)		(0.1711)	(0.1713)
Leverage		-0.2063***	-0.2060***		-0.1464	-0.1462
		(0.0506)	(0.0502)		(0.0914)	(0.0917)
Ind Num Takeover		0.7219***	0.7213***		0.7414***	0.7297***
		(0.1033)	(0.1019)		(0.1196)	(0.1184)
Year FE	Yes	Yes	Yes	No	No	No
Industry FE	Yes	Yes	Yes	No	No	No
Observations	91,488	91,488	91,488	91,547	91,547	91,547
Pseudo R-squared	0.0592	0.1120	0.1120	0.0164	0.0899	0.0900

Notes: This table reports coefficients from probit regressions of acquisition likelihood and the number of accretive targets. The dependent variable equals one if the firm is an acquirer and zero otherwise. Columns (1)–(3) include separate year and industry fixed effects; Columns (4)–(6) exclude them. Peers are defined by two-digit SIC industry, and the acquisition premium is 20%. Columns (2) and (5) include Ind P/FE Interdecile; Columns (3) and (6) include Ind P/FE SD. Covariates are lagged one calendar year. Continuous regressors use annual CDFs except the stated logarithmic transformation. The natural logarithm is applied to one plus the raw Accretive Targets count; zero counts are retained. The sample uses available observations from 1980-2022. Standard errors clustered by two-digit SIC industry are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table A5
Takeover Likelihood: Robustness Tests

	(1) 10%	(2) 30%	(3) 40%	(4) FF48	(5) Negative EPS	(6) Industry-year clusters
Accretive Bidders	0.2946***	0.2763**	0.2810**	0.3242***	0.3118***	0.2867***
	(0.1036)	(0.1230)	(0.1348)	(0.0922)	(0.0592)	(0.0894)
Ind P/FE Interdecile	-0.0677	-0.0731	-0.0773	-0.0785	-0.0692	-0.0708
	(0.0661)	(0.0704)	(0.0733)	(0.0591)	(0.0565)	(0.0493)
Ind Num Firms	-0.1718	-0.1713	-0.1734	-0.1815	-0.2467*	-0.1723
	(0.1268)	(0.1359)	(0.1410)	(0.1200)	(0.1414)	(0.1141)
P/FE	-0.3276***	-0.3334***	-0.3296***	-0.2952***	-0.4172***	-0.3291***
	(0.0806)	(0.0837)	(0.0872)	(0.0963)	(0.0503)	(0.0553)
Size	0.3120**	0.3181**	0.3197**	0.3919**	0.3823***	0.3156***
	(0.1363)	(0.1371)	(0.1375)	(0.1556)	(0.1286)	(0.1174)
Size Square	-0.6748***	-0.6801***	-0.6816***	-0.7235***	-0.7017***	-0.6779***
	(0.1209)	(0.1220)	(0.1224)	(0.1397)	(0.1203)	(0.1134)
Leverage	0.0915**	0.0905**	0.0902**	0.0808	0.0888*	0.0909***
	(0.0453)	(0.0454)	(0.0454)	(0.0527)	(0.0481)	(0.0346)
Ind Num Takeover	0.3536***	0.3582***	0.3589***	0.2804***	0.3847***	0.3562***
	(0.0827)	(0.0829)	(0.0830)	(0.0690)	(0.0834)	(0.0783)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	91,520	91,520	91,520	90,777	104,784	91,520
Pseudo R-squared	0.0542	0.0542	0.0542	0.0531	0.0509	0.0542

Notes: This table reports coefficients from probit regressions of takeover likelihood and the number of viable bidders. The dependent variable equals one if the firm is a takeover target and zero otherwise. All columns include separate year and industry fixed effects. Columns (1)–(3) use acquisition premiums of 10%, 30%, and 40%, respectively. Column (4) uses Fama–French 48 industries. Column (5) retains negative forecast EPS. Column (6) clusters standard errors by industry-year. The premium is 20% in Columns (4)–(6); peers are defined by two-digit SIC industry except in Column (4). Covariates are lagged one calendar year. Continuous regressors use annual CDFs. The sample uses available observations from 1980-2022. Standard errors are reported in parentheses and clustered by industry in Columns (1)–(5). $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table A6
Accretive, Book, and RIM Bidders: Robustness Tests

	(1) 10%	(2) 30%	(3) 40%	(4) FF48	(5) Negative EPS	(6) Industry-year clusters
Accretive Bidders	0.3520**	0.3749**	0.3866**	0.3236*	0.5070***	0.3751***
	(0.1701)	(0.1714)	(0.1804)	(0.1875)	(0.0982)	(0.1398)
Book Bidders	0.1784	0.1633	0.1482	0.2461**	0.1899	0.1822*
	(0.1412)	(0.1140)	(0.1042)	(0.0963)	(0.1223)	(0.1098)
RIM Bidders	0.2138***	0.1988***	0.1916***	0.1689**	0.2162***	0.2071**
	(0.0662)	(0.0686)	(0.0676)	(0.0805)	(0.0623)	(0.0921)
Ind P/FE Interdecile	-0.1148	-0.1326	-0.1409	-0.0123	-0.1353*	-0.1255**
	(0.0799)	(0.0864)	(0.0905)	(0.0818)	(0.0766)	(0.0628)
Ind Num Firms	-0.2120	-0.2159	-0.2133	-0.3178*	-0.3580**	-0.2217
	(0.1844)	(0.1915)	(0.1969)	(0.1812)	(0.1591)	(0.1478)
P/FE	-0.5924***	-0.5772***	-0.5717***	-0.6175***	-0.6376***	-0.5770***
	(0.1302)	(0.1280)	(0.1300)	(0.1306)	(0.0938)	(0.0798)
Firm Specific Error2	0.3569***	0.3562***	0.3507***	0.4359***	0.3670***	0.3626***
	(0.0898)	(0.0843)	(0.0812)	(0.0728)	(0.0861)	(0.0680)
Time Series Error2	0.0758	0.0756	0.0754	0.0606	0.0803*	0.0765*
	(0.0461)	(0.0487)	(0.0495)	(0.0512)	(0.0469)	(0.0396)
Long Run VtoB2	-0.1142**	-0.1173**	-0.1216**	-0.1333**	-0.0944*	-0.1118*
	(0.0541)	(0.0490)	(0.0477)	(0.0565)	(0.0509)	(0.0604)
Size	0.4232***	0.4362***	0.4416***	0.5360***	0.5559***	0.4294***
	(0.1552)	(0.1539)	(0.1535)	(0.1759)	(0.1669)	(0.1499)
Size Square	-0.8278***	-0.8396***	-0.8452***	-0.9141***	-0.9184***	-0.8330***
	(0.1391)	(0.1385)	(0.1387)	(0.1639)	(0.1560)	(0.1410)
Leverage	0.0591	0.0566	0.0558	0.0551	0.0554	0.0575
	(0.0592)	(0.0593)	(0.0593)	(0.0635)	(0.0623)	(0.0447)
Ind Num Takeover	0.3191***	0.3277***	0.3312***	0.2043**	0.2984***	0.3225***
	(0.1108)	(0.1107)	(0.1109)	(0.0837)	(0.1072)	(0.0936)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	62,634	62,634	62,634	62,269	63,120	62,634
Pseudo R-squared	0.0693	0.0693	0.0693	0.0676	0.0694	0.0694

Notes: This table reports coefficients from probit regressions of takeover likelihood and the number of viable bidders. The dependent variable equals one if the firm is a takeover target and zero otherwise. All columns include separate year and industry fixed effects. Columns (1)–(3) use acquisition premiums of 10%, 30%, and 40%, respectively. Column (4) uses Fama–French 48 industries. Column (5) retains negative forecast EPS. Column (6) clusters standard errors by industry-year. The premium is 20% in Columns (4)–(6); peers are defined by two-digit SIC industry except in Column (4). Covariates are lagged one calendar year. Continuous regressors use annual CDFs. The sample uses available observations from 1980-2022. Standard errors are reported in parentheses and clustered by industry in Columns (1)–(5). $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table A7
Marginal Effects of Accretive Targets at Different Levels of PPS

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Average marginal effect of Accretive Targets (percentage points per unit of its CDF)						
PPS CDF = 0.25	-2.017	-0.307	-0.311	-1.966	-0.884	-1.002
	[-4.50, 0.46]	[-2.66, 2.04]	[-2.65, 2.03]	[-4.45, 0.52]	[-3.22, 1.46]	[-3.44, 1.43]
PPS CDF = 0.50	-0.970	0.762	0.744	-1.037	0.039	-0.092
	[-3.32, 1.38]	[-1.54, 3.07]	[-1.56, 3.05]	[-3.55, 1.47]	[-2.35, 2.43]	[-2.65, 2.47]
PPS CDF = 0.75	0.028	1.861	1.829	-0.155	0.957	0.816
	[-2.50, 2.56]	[-0.82, 4.54]	[-0.86, 4.52]	[-2.98, 2.67]	[-1.84, 3.75]	[-2.23, 3.86]
Panel B: Change in probability as Accretive Targets CDF rises from 0.25 to 0.75 (pct. points)						
PPS CDF = 0.25	-1.019	-0.154	-0.156	-0.992	-0.443	-0.503
	[-2.28, 0.25]	[-1.33, 1.02]	[-1.33, 1.02]	[-2.26, 0.27]	[-1.62, 0.73]	[-1.73, 0.72]
PPS CDF = 0.50	-0.488	0.380	0.371	-0.521	0.019	-0.046
	[-1.68, 0.70]	[-0.77, 1.53]	[-0.78, 1.52]	[-1.79, 0.75]	[-1.17, 1.21]	[-1.33, 1.24]
PPS CDF = 0.75	0.014	0.924	0.908	-0.078	0.477	0.407
	[-1.25, 1.28]	[-0.40, 2.24]	[-0.42, 2.23]	[-1.49, 1.34]	[-0.91, 1.86]	[-1.10, 1.92]
Year fixed effects	Yes	Yes	Yes	No	No	No
Industry fixed effects	Yes	Yes	Yes	No	No	No
Observations	24,421	24,421	24,421	24,742	24,742	24,742
Pseudo R-squared	0.0737	0.1230	0.1225	0.0282	0.0704	0.0695

Notes: This table reports probability effects implied by the six probit specifications in Table 10. The dependent variable equals one if the firm is an acquirer and zero otherwise. Panel A reports the average derivative of acquisition probability with respect to the Accretive Targets CDF, in percentage points per unit of that CDF, with PPS incentive set to the stated value for every observation and other covariates, including Accretive Targets, at their observed values. Multiplying Panel A by 0.10 gives a local effect for a ten-percentile-point increase. Panel B reports the exact average change in probability when the Accretive Targets CDF rises from 0.25 to 0.75. The difference between the effects at PPS CDF = 0.75 and 0.25 is significant in Columns (2) and (3) ($p = 0.037$ and 0.040). Brackets report 95% confidence intervals using the delta method and standard errors clustered by two-digit SIC industry. Columns (1)–(3) include separate year and industry fixed effects; Columns (4)–(6) exclude them. Columns (2) and (5) include Industry P/FE SD; Columns (3) and (6) include Industry P/FE interdecile. The sample covers 1997–2022.

Table A8
Takeover Likelihood in the PPS-Covered Sample

	(1)	(2)	(3)	(4)	(5)	(6)
Accretive Bidders	0.5447***	0.1693	0.2064	0.3396***	0.2161	0.1851
	(0.0973)	(0.2363)	(0.2354)	(0.0731)	(0.1471)	(0.1687)
Ind P/FE SD		-0.1272			0.0732	
		(0.1082)			(0.0942)	
Ind P/FE Interdecile			-0.1943			0.1012
			(0.1189)			(0.1062)
Ind Num Firms		-1.0040***	-1.0581***		-0.3432*	-0.3186
		(0.3503)	(0.3691)		(0.2084)	(0.2157)
P/FE		-0.3120*	-0.2814*		-0.2536**	-0.2841**
		(0.1748)	(0.1706)		(0.1285)	(0.1418)
Size		0.4319	0.4146		0.3438	0.3520
		(0.2726)	(0.2746)		(0.3077)	(0.3080)
Size Square		-0.6983***	-0.6839***		-0.3927	-0.3931
		(0.2347)	(0.2352)		(0.2645)	(0.2661)
Leverage		0.1642*	0.1708*		0.1441	0.1357
		(0.0994)	(0.1002)		(0.0909)	(0.0888)
Ind Num Takeover		0.2436	0.2247		0.4041**	0.4080**
		(0.1799)	(0.1814)		(0.1927)	(0.1971)
Year fixed effects	Yes	Yes	Yes	No	No	No
Industry fixed effects	Yes	Yes	Yes	No	No	No
Observations	24,348	24,348	24,348	24,742	24,742	24,742
Pseudo R-squared	0.0497	0.0566	0.0568	0.0049	0.0095	0.0097

Notes: This table repeats the six specifications in Table 5 using firm-years with observed prior-year PPS5, the same PPS-coverage restriction used in Table 10. The dependent variable equals one if the firm is a takeover target and zero otherwise. Entries are probit index coefficients. Columns (1)–(3) include separate year and industry fixed effects; Columns (4)–(6) exclude them. Columns (2) and (5) include Ind P/FE SD; Columns (3) and (6) include Ind P/FE Interdecile. All controls, calendar lags, two-digit SIC peers and the 20% premium are unchanged. Continuous regressors retain their annual CDFs computed on the full baseline sample; they are not reranked within the PPS sample. The fixed-effect models exclude 394 firm-years in industries with no targets. The sample covers 1997–2022. Standard errors clustered by two-digit SIC industry are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Appendix B:
A Simple Model and Resulting Hypotheses

We develop a simple model of mergers following Shleifer and Vishny (2003) (hereafter SV) and then derive two new, empirically-testable predictions. As in SV, consider a representative merger pair, denoted firm 0 and firm 1. Firm 0 (firm 1) has K_0 (K_1) units of capital and the stock price is a multiple Q_0 (Q_1) of the capital. Assume without loss of generality that

$$Q_1 > Q_0,$$

so firm 1 is the prospective acquirer and firm 0 the prospective target. The key parameter in SV is *s*, the synergy that the market attaches to the combined entity. Specifically, the market value of the combined entity is

$$(K_1 + K_0)\,[sQ_1 + (1 - s)Q_0].$$

SV refer to this as the short-term market value, so that *s* can contain pricing errors. In the baseline case where there are no synergies and the market is efficient, $s = K_1/(K_1 + K_0)$. The target firm shareholders and management are assumed able to cash out immediately after the deal closes, so they are not concerned with longer-term value. Hence, what matters for the viability of the combination is the bidding firm's view of *s*.

We now impose two key components in addition to the SV model for our purpose of predicting the *incidence* of mergers. The first component of this analysis is the condition that the bidder must pay a non-zero premium given by a percent of target value. To illustrate it, assume without loss of generality that both firms have a single share outstanding and that the acquirer issues an additional *m* shares to the target. We now have two conditions for a merger pair to be viable. First, the bidder must provide enough shares, *m*, to cover the required premium of Π:

$$\frac{m}{1+m}[sQ_1 + (1 - s)Q_0](K_1 + K_0) = Q_0K_0(1 + \Pi). \quad \text{(B1)}$$

Second, the bidder must not expect to lose market value from the bid:

$$\frac{1}{1+m}[sQ_1 + (1 - s)Q_0](K_1 + K_0) \geq Q_1K_1. \quad \text{(B2)}$$

Conditions (B1) and (B2) are satisfied so long as:

$$\frac{Q_1 - Q_0}{Q_0} > \frac{\Pi K_0}{s(K_1+K_0) - K_1}. \quad \text{(B3)}$$

In the data we cannot directly observe the bidder's beliefs about *s*. Two extreme cases are instructive. In an efficient market and management that doesn't believe in synergy (i.e., the case where $s = K_1/(K_1 + K_0)$), then the two conditions above can never be satisfied for any $\Pi > 0$.

This simply says that the bidder cannot offer a premium if there is no gain to the merger, efficiency-based or otherwise. At the other extreme, if *s* = 1 (the bidder at least believes the market will apply her higher multiple to the target's assets), then all that is required is for the bidder's multiple to exceed that of the target by the premium. This is a straightforward, if extreme, bootstrapping result.

When a pair of firms satisfy the above conditions, they are a viable merger pair. However, as we can see later, we can identify many viable pairs from the data while the takeover events are much less frequent. There are many other dimensions which will prevent a merger from happening in the reality. Hence, taking our highly simplified model to the real-world data requires us to consider (at least) three issues.

First, for most reasonable premia, the range of multiples in broadly-defined industries implies that many firms have multiple viable buyers; this implies an unconditional likelihood of a takeover at least an order of magnitude greater than what we observe in the data. There are several reasons to accommodate this fact that most viable buyers will not bid. One simple reason is that some viable bidders do not believe in the bootstrap game. The potential integration cost and risk of acquisition can also keep some viable buyers from pulling the trigger. For example, it will be challenging for an acquirer without unionized workers to integrate a target with unionized workers. The failure of mergers due to the cultural difference is not uncommon. Also, the search literature on mergers suggests that bidders and targets face a search friction and only find each other with some probability (see Rhodes-Kropf and Robinson 2008). Hence, a fraction *X* of the population will not become an actual bidder. Since only one viable bidder is required for the firm to be taken over, if we denote the number of firms that satisfy conditions (B1) and (B2) by *n*, a potential target is taken over with probability $1 - X^n$. While we do not actually know *X*, this observation suggests we should apply a concave transformation to the number of viable bidders when we come to our empirical tests.[14]

Second, relative size does not matter, because when *s* = 1 even a small bidder can apply her multiple to a large target. This may seem counterintuitive, but Harford (1999) surprisingly finds that targets are *not*, on average, small firms and we confirm this basic result in our data. To further investigate the idea, in robustness tests we add the requirement that the bidder's assets be greater than those of the target and our results hold.

Finally, we have thus far adopted the Shleifer-Vishny modeling assumption that differences in multiples are primarily due to mispricing, rather than appropriate valuation of differential risks or cash flow expectations (at least, we have done so in the *s* = 1 case). Much of the subsequent empirical literature (Dong et al. 2006, Rhodes-Kropf et al. 2005) attempts to specify

[14] This is particularly the case if we were to endogenize the required premium; it is likely to increase in the number of viable bidders and accentuate the diminishing-returns effect.

valuation models in order to isolate mispricing. Our approach hinges more on bidder behavior and beliefs than any specific pricing model. For the target, the deal is acceptable simply if it offers a sufficient premium. For the buyer, so long as the deal can increase its per share value (either EPS, Book Value, or Intrinsic Value), the deal is viable. We then argue that the likelihood of a merger is related to the number of viable buyers or targets. For a firm to sell itself, the possibility for it to find a buyer that is able to pay it a desired premium increase with the number of viable buyers. The more viable buyers, the larger the chance that at least one buyer is able to pay target's desired premium.

Appendix C: SDC Data Details

We drop records without announcement date and the duplicate records when all of the target's CUSIP, acquirer's CUSIP, date announced, and status are the same, leaving us with an initial sample of 315,603 records of announcements. We also drop deals for which the target's CUSIP or target's parent's CUSIP equals to either of the acquirer's CUSIP or acquirer's parent's CUSIP. We further drop the record if the same target and acquirer pair was recorded more than once in the same year. This gives us a raw sample of 309,003 records.

We then require both targets and acquirers to be public and are left with 12,448 records. We then combine this SDC data with a file called "stocknames" from WRDS in order to attach eight-digit CUSIP to the targets. Then, we merge with 'stocknames' file again to get eight-digit CUSIP for the acquirors. The linking file identifies targets and acquirors by PERMNO. We further drop the record if the same target was reported more than once in the same year. After applying these filters, we have a sample of 11,979 records and each target has only one record in the same calendar year. To merge with CRSP Compustat, we obtain the calendar year from CRSP Compustat (the date is the day on the end of the fiscal year up to which the company reports it annual statement). Then, we merge SDC with CRSP Compustat by matching PERMNO and calendar year. This step matches 4,077 target firms into CRSP Compustat and we create a dummy variable, *takeover*, which takes the value of 1 for the target in the calendar year.

The above combination process misses many target firms because some do not provide annual reports in the year in which they were acquired. For those firms, we match the announcement to the preceding calendar-year record in the CRSP Compustat and then merge the data again. The second step matches 4,308 takeover records into the CRSP Compustat dataset. Of these, 108 overlap with the first match, so this step adds 4,200 records, leaving us with a total of 8,277 records before the acquirer match. For the acquirer sample, it is relatively easy because usually the acquirers' financial data are available in the merger year. We also restrict this sample to 1979 to 2022, resulting in 9,281 acquirer records. The number is a little larger than that of targets alone because more financial data are available for the acquirers and some targets received multiple bids. After combining the matches and assigning acquirer status when an acquirer match is present, the linked data contain 7,926 target records and 9,281 acquirer records. These correspond to 7,925 distinct target firm-years and 8,251 distinct acquirer firm-years.

We retain one observation for each firm and calendar year. The constructed panel contains 271,744 distinct firm-years from 1979 to 2022. We require the previous calendar year's analyst EPS forecast and stock price, leaving 106,905 firm-years, including 4,780 targets and 4,257 acquirers. We exclude observations with negative or zero forecast EPS, leaving 93,136 firm-years, including 4,044 targets and 4,054 acquirers. Requiring eligible accretive-bidder and accretive-target counts leaves 93,023 firm-years. Our final sample has 91,547 firm-years from 1980 to

2022: 3,959 targets, 3,992 acquirers and 83,596 non-mergers. These are firm-year observations, so a firm can appear in more than one year.

The reduction from the linked data reflects the analyst forecast requirement, the positive forecast-EPS restriction, valid prior-year accretion counts and the availability of the required controls.[15]

In the baseline Accretive Bidders and Accretive Targets measures, both the focal firm and the potential counterparty must have positive forecast EPS, a positive stock price and positive shares outstanding. Negative and zero forecast EPS are excluded from the counts. Missing forecasts are not filled with reported EPS in the baseline sample. Appendix Tables A5 and A6 retain negative forecast EPS as a separate robustness check. These rules concern forecast EPS.

Form of the Transaction: Codes used in sampling M&A transactions include four forms:

M (MERGER): A combination of business takes place or 100% of the stock of a public or private company is acquired.

A (ACQUISITION): deal in which 100% of a company is spun off or split off is classified as an acquisition by shareholders.

AM (ACQ OF MAJORITY INTEREST): the acquiror must have held less than 50% and be seeking to acquire 50% or more, but less than 100% of the target company's stock.

Acquisition of Assets: A transaction involving the purchase of a company's assets.

Transaction Type Code: Code number for the type of transaction (e.g. 1=DI):

1 = Disclosed Value: indicates all deals that have a disclosed dollar value and the acquiror is acquiring an interest of 50% or over in a target, raising its interest from below 50% to above 50%, or acquiring the remaining interest it does not already own.

2 = Undisclosed Value: indicates all deals that do not have a disclosed dollar value and the acquiror is acquiring an interest of 50% or over in a target, raising its interest from below 50% to above 50%, or acquiring the remaining interest it does not already own.

3 = Leveraged Buyouts: indicates that the transaction is a leveraged buyout. An "LBO" occurs when an investor group, investor, or firm offers to acquire a company, taking on an extraordinary amount of debt, with plans to repay it with funds generated from the company or with revenue earned by selling off the newly acquired company's assets. TF considers a deal an LBO if the investor group includes management or the transaction is identified as such in the financial press and 100% of the company is acquired.

4 = Tender Offers: indicates a tender offer is launched for the target. A tender offer is a formal offer of determined duration to acquire a public company's shares made to equity

[15] Of the distinct target firm-years in the constructed input panel, the final sample retains 40.6% in the 1980s, 48.7% in the 1990s, 50.1% in the 2000s, 61.3% in the 2010s and 56.6% in 2020–2022. The corresponding acquirer percentages are 35.4%, 41.7%, 52.2%, 66.4% and 73.4%. No 1979 observation enters the analysis because prior-year inputs are required.

holders. The offer is often conditioned upon certain requirements such as a minimum number of shares being tendered.

5 = Spinoffs: indicates a "spinoff," which is the tax-free distribution of shares by a company of a unit, subsidiary, division, or another company's stock, or any portion thereof, to its shareholders. TF tracks spinoffs of any percentage.

6 = Recapitalizations: indicates a deal is a recapitalization, or deal is part of a recapitalization plan, in which the company issues a special one-time dividend in the form of cash, debt securities, preferred stock, or assets, while allowing shareholders to retain an equity interest in the company.

7 = Self-Tenders: indicates all deals in which a company announces a self-tender offer, recapitalization, or exchange offer. In a self-tender offer, a company offers to buy back its equity securities or securities convertible into equity through a tender offer. A company essentially launches a tender offer on itself to buy back shares.

8 = Exchange Offers: indicates a deal where a public company offers to exchange new securities for its outstanding securities. Only those offers seeking to exchange consideration for equity, or securities convertible into equity, are covered in the M&A database. See EXCHANGE OFFER DATABASE for transactions involving debt. 9 = Repurchases: indicates all deals in which a company buys back its shares in the open market or in privately negotiated transactions or a company's board authorizes the repurchase of a portion of its shares.

10 = SP: indicates all deals in which a company is acquiring a minority stake (i.e. up to 49.99% or from 50.1% to 99.9%) in the target company.

11 =Acquisitions of Remaining Interest: indicates all deals in which a company is acquiring the remaining minority stake (i.e. from at least 50.1% ownership to 100% ownership), which it did not already own, in a target company. The acquiring company must have already owned at least 50.1% of the target company and would own 100% of the target company at completion.

12 = Privatizations: indicates a government or government-controlled entity sells shares or assets to a non-government entity. Privatizations include both direct and indirect sales of up to a 100% stake to an identifiable buyer and floatations of stock on a stock exchange. The former is considered an M&A transaction and will be included in the quarterly rankings; the latter will not.

Appendix D: Empirical Variable Definitions

Table D1
Description of Key Variables

The variables used in the empirical analysis are defined as follows. Sources are reported in the final column.

Variable	Description	Source
Size	Is the natural logarithm of the book value of total assets, ln*(at)*.	Compustat
Leverage	Is the ratio of sum of long term and short term debt to total assets, *(dltt+dlc)/at*.	Compustat
P/FE	The ratio of stock price to the forecasted earnings per share, *prcc_c/MEANEST*. The baseline uses the one-year analyst forecast.	Compustat; I/B/E/S
Ind Num Takeover	Is the number of takeovers in the target's two digit SIC industry in a particular year. Each target firm-year is counted once.	SDC; author calculations
Ind Num Firms	Is the total number of firms in a two digit SIC industry each year. Firms are counted once per industry-year in the constructed universe, before the forecast and final regression-sample exclusions.	Compustat; author calculations
Ind P/FE Interdecile	Is the difference between the industry's 90th percentile P/FE ratio and the industry's 10th percentile P/FE ratio each year.	Compustat; I/B/E/S; author calculations
Ind P/FE SD	Is the standard deviation of the industry's P/FE ratio each year.	Compustat; I/B/E/S; author calculations
Market Value	Market value of the shares outstanding, *mkv=prcc_c×csho*.	Compustat
Market to Book	The ratio of the Market value of equity to the book value of equity, *mtb=mkv/ceq*.	Compustat
Book value Per Share	*bvps=ceq/csho*.	Compustat
S&P 500	The level of the Standard & Poor's 500 Composite Index, *spindx*. The series is the year-end level and is constant within calendar year; the aggregate regressions average that constant across firms.	Compustat
Dividend	An indicator variable for whether or not the firm pays a dividend. It equals one when common dividends are positive, zero otherwise when dividend data are observed, and missing when dividend data are missing.	Compustat
Regulated	An indicator variable equal to 1 if the firm is in the financial industry (SIC 6000–6999) or SIC 4800–4914, and 0 otherwise.	Compustat
Accretive Bidders	*B* is an accretive bidder for *T* if $x=1.2p_T/p_B$ is the all stock purchase exchange ratio and $e=(fe_Tn+fe_Bm)/(nx+m)>fe_B$. Here *n* and *m* are the number of common shares outstanding for the target and bidder respectively, and fe_T and fe_B are forecasted EPS. Accretive Bidders counts the distinct eligible bidders in the same industry-year, excluding the firm itself.	Compustat; I/B/E/S; author calculations
Accretive Targets	The number of viable targets there are for each bidder, using the same forecast-EPS accretion rule, industry-year universe and 20% premium as Accretive Bidders. Self-matches are excluded.	Compustat; I/B/E/S; author calculations

Continued on next page

Table D1 continued

Variable	Description	Source
Book Bidders	The number of book value based viable bidders for a firm. Replace forecast EPS in the accretion rule by book equity per share. Both firms must have positive book equity per share, price and shares outstanding. Count distinct firms in the same industry-year and exclude self-matches.	Compustat; author calculations
Book Targets	The number of book value based viable targets for a firm, using the same pair eligibility and 20% premium as Book Bidders.	Compustat; author calculations
RIM Bidders	The number of intrinsic value (calculated by residual income model) based viable bidders for a firm. Target *T* and bidder *B* satisfy $V_T p_B > 1.2 p_T V_B$. Both firms must satisfy the RIM eligibility screens below. Count distinct firms in the same industry-year and exclude self-matches. Negative intrinsic values remain eligible; the signed cross-product rule is used.	Compustat; I/B/E/S; CRSP; author calculations
RIM Targets	The number of intrinsic value based viable targets for a firm, using the same RIM pair eligibility and 20% premium as RIM Bidders. Bidder and target count totals are equal within each industry-year.	Compustat; I/B/E/S; CRSP; author calculations
Intrinsic value (V)	Three-year RIM value per share: $V=B_0+(ROE_1-r)B_0/(1+r)+(ROE_2-r)B_1/(1+r)^2+(ROE_3-r)B_2/[r(1+r)^2]$. Here $B_0=ceq/csho$, $B_h=B_{h-1}+(1-k)F_h$ for $h=1,2$, and F_h denotes the analyst forecast. Forecast ROEs use average book equity: $ROE_1=F_1/[(B_0+B_{-1})/2]$, $ROE_2=F_2/[(B_1+B_0)/2]$, and $ROE_3=F_3/[(B_2+B_1)/2]$. No additional split factors are applied.	Compustat; I/B/E/S; CRSP; author calculations
Dividend payout ratio (k)	Reported common dividends per share divided by reported EPS, *dvpsx_f/epspx*. Reported EPS must be positive and $0 \le k \le 1$. Missing dividends or reported EPS remain missing. There is no 6%-of-assets fallback.	Compustat
Cost of equity (r)	Firm-year CAPM rate bounded between 3% and 30%. Beta uses up to five years of monthly returns with at least 24 observations. The market premium uses the preceding 30 complete annual observations, and the risk-free rate is annualized from the preceding completed month.	CRSP; market and risk-free series; author calculations
P/B	The ratio of stock price to book value per share, *prcc_c/bvps*.	Compustat
P/V	The ratio of stock price to the RIM intrinsic value per share, *prcc_c/V*. Negative *V* remains eligible; P/V is not winsorized.	Compustat; I/B/E/S; CRSP; author calculations
Size Square	The square of the Size CDF in the CDF specifications. In the PPS and liquidity tables, it is the square of log assets, as specified in the table notes.	Author calculations
Log(Accretive Bidders); Log(Accretive Targets)	The natural logarithm of the corresponding raw lagged count. Observations with zero counts are excluded because the logarithm is undefined. The logged count enters directly, without a further CDF transformation.	Author calculations

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Table D1 continued

Variable	Description	Source
Log(1+Accretive Bidders); Log(1+Accretive Targets)	The natural logarithm of one plus the corresponding raw lagged count. Zero counts remain in the sample. The transformed count enters directly, without a further CDF transformation.	Author calculations
Firm Specific Error2	The difference between log market equity and fitted log value using the current industry-year coefficients: $m\text{-}v(\vartheta;\alpha_{jt})$. The suffix 2 identifies valuation Model 2; it does not mean the variable is squared.	Compustat; author calculations
Time Series Error2	The difference between fitted log value using current industry-year coefficients and fitted log value using long-run industry coefficients: $v(\vartheta;\alpha_{jt})\text{-}v(\vartheta;\bar{\alpha}_j)$.	Compustat; author calculations
Long Run VtoB2	The difference between fitted log value using long-run industry coefficients and log book equity: $v(\vartheta;\bar{\alpha}_j)\text{-}b$. Together, the three Model 2 components add to $m\text{-}b$.	Compustat; author calculations
Ind Average Time Series Error	The mean of available raw lagged Time Series Error2 values across retained firms in a two-digit SIC industry and year. It enters Table 4 without a CDF transformation. The underlying valuation model is estimated within Fama–French 12 industries; aggregation here uses two-digit SIC.	Compustat; author calculations
Ind Average long-run VtoB	The mean of available raw lagged Long Run VtoB2 values across retained firms in a two-digit SIC industry and year. It enters Table 4 without a CDF transformation and uses the same valuation model as Time Series Error2.	Compustat; author calculations
PPS	The earnings-based pay-performance sensitivity estimate from Bennett et al. (2017), *b_eps_con_ret_5* (PPS5), used to construct the focal firm's own PPS incentive measure.	Incentive Lab
PPS incentive	The empirical CDF of the focal firm's own PPS5 across all distinct firms with observed PPS5 in the source year, before EPS eligibility and regression-sample restrictions. It equals the share with PPS5 no greater than the focal value; ties receive the same CDF. Negative and zero PPS are retained; missing PPS remains missing. Lagged one exact calendar year in Table 10.	Incentive Lab; author calculations
Accretive Targets × PPS incentive	The product of the annual Accretive Targets CDF and the focal firm's own lagged PPS5 CDF. Table 10 includes the interaction and both main effects in probit regressions of the acquirer dummy.	Author calculations
Cash ratio	Cash divided by total assets, *ch/at*.	Compustat
Liquid accretive bidders	The number of distinct EPS-accretive bidders with cash/assets strictly above the 75th percentile across all firms with observed cash/assets in the same source year. The cutoff uses unique firm-years and linear interpolation before EPS eligibility restrictions. Ties and missing cash do not qualify. Bidders remain in the focal firm's two-digit SIC industry; exclude self-matches and duplicates. The count is lagged one calendar year and enters as its annual empirical CDF, calculated on the full baseline reference population before model exclusions, alongside the annual CDF of total Accretive Bidders.	Compustat; I/B/E/S; author calculations
Target dummy	An indicator variable equal to 1 if a firm is a takeover target and 0 otherwise. The saved SDC linkage assigns acquirer status first if an acquirer match is present.	SDC; CRSP linkage
Acquirer dummy	An indicator variable equal to 1 if a firm is an acquirer and 0 otherwise. Multiple deals involving the same firm in a year contribute only one firm-year observation.	SDC; CRSP linkage
Merger dummy	An indicator variable equal to 1 if a firm is a takeover target or an acquirer and 0 otherwise.	SDC; CRSP linkage

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Table D1 continued

Variable	Description	Source
Merger rate	The number of distinct target firm-years divided by the number of distinct firms in the industry-year input universe. Panel D of Table 1 correlates this rate with industry-year medians of the lagged count measures, repeated across retained firm-years; each retained firm-year receives equal weight.	SDC; Compustat; author calculations
Median accretive bidders; Industry mean accretive bidders	The median across firms in a calendar year, or the mean across firms in an industry-year, respectively. Current and lagged values are identified in Tables 3 and 4.	Author calculations
Market-to-book interdecile	The difference between the industry's 90th and 10th percentile market-to-book ratios each year. The annual regressions use the median of this dispersion measure across retained firm-years.	Compustat; author calculations
Pre-period	Equals 1 in 1991–2000 for the ten-year window or 1989–2000 for the twelve-year window, and 0 in the respective post-periods 2003–2012 or 2003–2014. Outcome years 2001 and 2002 are excluded. With year fixed effects, the standalone indicator is absorbed.	Author calculations
Pre-period × accretive bidders	The product of the pre-period indicator and the annual Accretive Bidders CDF. It enters the combined regressions in Table 9.	Author calculations
Size, Dividend or Regulated × accretive targets	The product of the Size CDF or the indicated dummy and the annual Accretive Targets CDF, as labeled in Table 8.	Author calculations